

Cost Dynamics in Healthcare Provider Markets: An Empirical Study of Hospice Care

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Abstract

In the California hospice industry, per-visit costs are lower in markets with greater competition, due to the following. For-profits have lower costs than non-profits, and substantial for-profit entry has driven down the distribution of costs. New entrants have lower per-visit costs than incumbents and prior entrants, and hospices with higher per-visit costs are more likely to exit. However, incumbents facing higher competition do not reduce per-visit costs, and greater competition does not force higher-cost incumbents to exit. I find no evidence that lower cost reduces quality.

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1 Introduction

What is the effect of industry dynamics on cost efficiency in healthcare provider markets? Such industry dynamics may lead to high-cost firms exiting, low-cost firms entering, and incumbent firms being forced to reduce costs to stay competitive. However, there is little empirical evidence on this question for healthcare provider markets. This issue is further complicated by for-profit firms competing with non-profits in the same market. Answering this question will shed light on how competition may improve economic efficiency and help reduce rising healthcare costs in the United States.

I study how per-visit costs of hospices in California evolve with entry-exit dynamics. Hospices primarily serve patients by visiting them at their residences, so I focus on a hospice's average cost of making a visit, i.e. per-visit cost. Per-visit costs are lower in markets with more hospices, and I find this pattern to be driven by the following. For-profits have lower per-visit costs than non-profits, new entrants have lower per-visit costs than incumbents as well as prior entrants, and hospices with higher per-visit costs are more likely to exit. Incumbents facing higher competition do not lower their per-visit costs, and greater competition does not force higher-cost incumbents to exit. These results hold even after controlling for patient characteristics that may influence visit costs. I also do not find any evidence that lowering costs leads to lower quality.

This informs policy discussions on entry barriers and for-profit competition in healthcare markets. Provider markets in the US and elsewhere face a wide range of entry restrictions, such as licensing restrictions, scope-of-practice laws, and Certificate-of-Need (CON) laws. Furthermore, entry by for-profits is viewed with suspicion in political circles and the media, and prior research has found that for-profits provide lower quality care in some settings. More specifically to California, the hospice industry has recently been a controversial topic and seen newly-erected entry barriers. The last few years have seen multiple reports from the California legislature, state auditor, and California Department of Justice about fraud committed by hospices in Los Angeles. This has culminated in a moratorium on new hospice entry in California through CON laws, first imposed

in 2021 and recently extended through to 2027 and possibly beyond.¹

Hospice staff provide palliative care to terminally ill patients by visiting them at their residence. Most hospice patients are covered by Medicare and Medi-Cal, so hospices do not compete on prices. I use a panel dataset of hospices in California for 2002-2018 containing information about visits, costs, patient volume, and other firm-level statistics. I exclude Los Angeles from my main analysis due to recent allegations of fraud, but present results including LA in the Appendix.

First, I present descriptive results about the California hospice industry. I find that the number of hospices in California has grown tremendously between 2002-2018, and that this growth is mostly driven by entry of for-profit hospices. For-profits have improved patient access by serving many more patients than non-profits, while non-profits (aided by donations) have improved patient access by operating in smaller markets eschewed by for-profits. Hospices in more competitive markets visit their patients more frequently, suggesting increased quality of care. Per-visit costs are highly persistent, which when coupled with the importance of visit costs in overall hospice costs, suggests that it may play an important role in market dynamics.

Using linear and quantile regressions with instrumental variables, I show that per-visit costs are lower in counties with greater competition (i.e. more hospices). I regress per-visit costs on number of hospices to show that a negative relationship exists. A confounder is that firms may be entering counties which have exogenously lower costs. To rule that out, I use market size (measured as the expected number of deaths in a market) to instrument for number of hospices, and also control for market-level cost trends flexibly.² I also use a grouped-quantile IV regression (Backus (2020), Larsen et al. (2023)) to show that the distribution of per-visit costs shifts left with greater competition.

Why does competition lower per-visit costs? I argue that the negative coefficient on com-

¹See Section J for discussion and links to reports from the California state auditor and Department of Justice, as well as news outlets. The main type of fraud seems to be people setting up fake hospices to bill Medicare without providing services. The concern for fraud appears to be localized in Los Angeles according to the state auditor's investigation. While there have been accusations of widespread fraud in LA, the actual extent of it is unclear, as there have been relatively few prosecutions. However the investigations are ongoing at the time of this writing.

²Using demand shifters as instruments for competition level is motivated by Backus (2020), among others. The intuition is that market size shifts the number of firms operating in a market (the endogenous variable) without affecting their productivity or per-visit costs (the outcome) directly.

petition could comprise three mechanisms: greater competition occurs through entry of lower-cost firms (entry-composition effect), greater competition forces higher-cost firms to exit the market (exit-selection effect), or greater competition forces incumbents to lower their per-visit costs (incumbent-treatment effect).

I examine which of these mechanisms are driving the negative relationship between per-visit costs and competition. I find motivating evidence (via graphical methods and descriptive regressions) that for-profits have lower per-visit costs than non-profits, and that later entrants have lower per-visit costs than earlier entrants. These provide strong suggestive evidence for the entry-composition effect, especially as the growth in hospices in California is primarily from for-profit entry. I also find that higher-cost incumbents are more likely to exit the market, although this does not appear to be driven by greater competition.

Therefore, I rerun the previous regression of per-visit costs on number of hospices, this time also controlling for whether the hospice is a for-profit and entry year of the hospice. Since these should be absorbing the entry-composition effect, the coefficient on competition level should now be the sum of exit-selection and incumbent-treatment effect; this is similar to the argument in [Backus \(2020\)](#). Strikingly, I find that the negative relationship between per-visit costs and number of hospices now disappears. I continue to find that for-profits and later entrants have lower costs. These therefore suggest that entry-composition effect is the primary mechanism driving the negative relationship between competition and per-visit costs. I perform an additional check to rule out the incumbent-treatment effect by seeing if hospices that entered prior to 2002 show any cost declines when facing greater competition, and find that they do not.

Motivated by the palliative care literature, I show that these results are robust to controlling for patient characteristics and hospice characteristics that may influence visit costs. Finally, I redo the analysis including Los Angeles county, and with restricting to only small markets, and find qualitatively same results.

I find no evidence that reduction in per-visit costs leads to a decline in the quality of care. In fact, hospices with lower per-visit costs tend to make more visits to patients, which should improve

patient experience. The cost reductions I document are not entirely driven by substituting cheaper home health aides for more expensive nurses. I also find evidence for reallocation of market share towards lower-cost providers in more competitive markets. Finally, I speculate about possible economic mechanisms that may be driving my results. Among others, I argue that increasing competition means only low-cost hospices can enter and make sufficient profits while still providing frequent visits to patients.

Therefore, this paper sheds light on provider entry without entry restrictions in general, and the California hospice industry in particular. The absence of entry barriers in the California hospice industry has meant that per-visit costs have declined in areas facing greater competition. This cost cutting allows Medicare to pay low reimbursement rate to hospices without risking driving them out of the market, and thus lowers healthcare expenditure. Patient access to care has improved significantly due to for-profit entry, with for-profits now serving more patients than non-profits. Markets with greater hospice competition also see more frequent visits to patients, suggesting quality improvements. Thus, a key policy insight is that entry barriers – such as the recently implemented CON laws for hospices in California – can i) impede cost reductions and quality improvements from competition, as well as ii) worsen patient access to care, since for-profits have improved access for patients over the dataset’s period.

Contributions and related literature: First, I contribute to a small literature on how health-care providers’ efficiency and productivity evolve over time and with industry dynamics. My results connect to the general discussion in [Cochrane \(2015\)](#), who argues that competition in health-care markets can drive down costs by improving efficiency of providers as well as improve quality of care. My results also connect to [Chandra et al. \(2016\)](#), who push back against the idea of “healthcare exceptionalism”. Their paper argues that healthcare industry is similar to other industries in that higher quality hospitals see higher market share and growth. My paper contributes to this discussion by examining the cost side and showing that, similar to non-healthcare industries, increasing competition drives down costs over time.

In contrast to this paper, few papers in the Economics literature study the effect of competition

on healthcare provider costs, and none study this question in the context of hospices. An advantage of studying hospices is that hospice services are simple and low-skill compared to other healthcare providers. Thus, there is less concern about unobserved variation in type of services, patient case-mix, learning-by-doing, etc. Most papers studying competition in healthcare markets focus on prices and quality. Furthermore, papers studying healthcare provider costs and productivity often focus on hospitals, such as [Gaynor et al. \(2015\)](#).

Second, I contribute to the literature studying the effect of competition on efficiency, productivity, and reallocation. I do so by showing empirically that entry of low-cost producers play an important role in shifting average production costs across markets and over time. In contrast to most other papers in the literature, I focus on cost-based measures to study efficiency. The advantage is that this does not require me to make strong assumptions for estimation (such as those made in the production function estimation literature; see [Olley and Pakes \(1996\)](#) and subsequent literature), and such cost-based efficiency measures can be recovered directly from (often widely available) cost data. [Asker et al. \(2019\)](#) is another paper that uses accounting cost data to study efficiency and misallocation, in their case for the global oil industry. Increasingly, papers are leveraging cost data due to their richness and availability; see [Igami et al. \(2024\)](#). A key paper for my work is [Backus \(2020\)](#), who studies the effect of competition on pushing out low-productivity ready-mix concrete plants and forcing incumbents to become more productive; I make extensive use of concepts and techniques used in that paper. Other papers in this area include [Collard-Wexler \(2011\)](#), [Collard-Wexler and De Loecker \(2015\)](#), [Fabrizio et al. \(2007\)](#), [Syverson \(2004a\)](#), [Syverson \(2004b\)](#), [Galdón-Sánchez and Schmitz \(2002\)](#), [Schmitz \(2005\)](#), and [Demirer and Karaduman \(2025\)](#); see [Syverson \(2011\)](#) and [De Loecker and Syverson \(2021\)](#) for surveys.

Third, I contribute to a very small literature on hospices in the Economics literature. Despite the US hospice industry being large and growing – over 1.6 million Medicare patients used hospices in 2022, and Medicare spent around \$21 billion on hospice services in 2019 alone – there are few papers studying how this industry operates. I am the first to study the dynamics regarding entry, exit, and production costs in the hospice industry. I also combine insights from the palliative care

literature to capture the shape of and inputs into the hospice cost function. [Alam \(2025\)](#) uses a structural model of the California hospice industry to study how hospices compete on quality and reputation, then simulates the impact of alternative Medicare hospice reimbursement scheme.³ An important paper is [Chung and Sorensen \(2018\)](#), who look at broad patterns of hospice entry across the US, then estimate a nested logit demand model to examine market expansion from hospice entry. Other prior work include [Dalton and Bradford \(2019\)](#), who study length of stay of patients in non-profit and for-profit hospices; and [Gruber et al. \(2023\)](#), who show that hospices save money for Medicare by offsetting alternative expensive care for Alzheimer’s and Dementia patients.

Fourth, I contribute to the literature studying differences between for-profits and non-profits in two ways. First, I contribute by studying production cost differences between for-profit and non-profit hospices; very few papers in the Economics literature study cost differences between for-profit and non-profit healthcare providers. Second, I show that entry by for-profits (and lack thereof for non-profits) has led to for-profits serving significantly more patients than non-profits, thereby improving patient access to care. Several papers study and model entry decisions by non-profits versus for-profits in non-hospice settings; these include [Cohen et al. \(2013\)](#), [Gayle et al. \(2017\)](#), [Harrison and Seim \(2019\)](#), [Lakdawalla and Philipson \(2006\)](#), and [Philipson and Lakdawalla \(2001\)](#).

Fifth, I contribute to the literature on entry-exit dynamics in Empirical Industrial Organization.

³I expand more on the findings of [Alam \(2025\)](#) so as to distinguish its contributions from the current paper. It uses the same dataset as the current paper. It uses a structural demand model to show how hospices choose quality to accumulate reputation and get patients – this is effectively a dynamic form of quality competition. Then it estimates a simple cost function of hospice quality, recovering the economic cost of making a visit. It embeds these in a structural oligopoly model of reputation competition in the hospice industry, and uses it to solve for equilibrium hospice quality and simulate counterfactuals. The counterfactuals study how reputation effects and alternative reimbursement schemes can be used to improve equilibrium hospice quality and lower Medicare spending.

Generally the paper makes little use of the cost data (only for instruments in the demand model and to inform the cost function) and focuses on a subset of counties in California. The paper does not study hospice entry-exit, and assumes costs are homogeneous across hospices for computational tractability. The aim of the cost function is to recover the economic cost of making a visit, so as to make accurate predictions of quality choices in counterfactuals.

By contrast, the current paper focuses on the cost data of all counties in California (although LA is left out of the main analysis and included in the Appendix). It studies accounting cost heterogeneity across hospices, and how it varies across entrants, incumbents, and exiting firms.

In [Alam \(2025\)](#), two reduced form facts – higher competition leads to more visits-per-patient-day, and hospices with greater visits-per-patient-day have greater market share – are explored in great detail for a subset of counties. In the current paper, these facts are extended to all counties in California, and restated in the main text for important contextualization.

This paper brings reduced-form evidence that can inform structural modeling approaches in the literature. Most papers in this literature combine structural methods with parametric assumptions to impute economic costs of firms. Papers in this area using structural estimation methods include [Collard-Wexler \(2013\)](#), [Ericson and Pakes \(1995\)](#), [Igami \(2018\)](#), [Igami and Uetake \(2020\)](#), and [Ryan \(2012\)](#); see [Doraszelski and Pakes \(2007\)](#) and [Arcidiacono and Ellickson \(2011\)](#) for overview.

2 Industry Background and Data

Hospice staff provide palliative care to terminally ill patients by visiting them at their residence. Patients who enroll in hospice typically no longer receive curative treatment. Most patients in the dataset are covered by Medicare or Medi-Cal, which pays hospices a fixed amount for every day the patient is enrolled. In this paper, I use a panel dataset of hospices in California for 2002-2018 containing information about visits, costs, patient volume, and other firm-level statistics. This allows me to back out estimates of per-visit cost and competition levels in each market-year. I define a market to be at the county level, and justify this decision with data on patient origins. Market size is measured as the expected number of deaths in a market. Parts of this section closely follow an earlier paper by the author ([Alam \(2025\)](#)).

2.1 Hospice care provision

Hospices typically provide care at the residence of the patient⁴. Hospices employ several types of staff: registered nurses, licensed vocational nurses, home healthcare aides, physicians, social workers and chaplains. Nurses and home health aides make the majority of hospice visits.

The hospice staff make multiple visits to a patient throughout the patient’s enrollment period.

⁴My dataset (and Medicare) distinguishes between different types of hospice care – routine care (which is the care described above), inpatient care (where a patient moves to an inpatient ward of a hospice for intense treatment) and continuous respite care (where the hospice takes over as the primary caregiver). The majority of patients in my sample are given routine care; I show in Table 23 that of all the “days of care” provided by a hospice within each year, over 99% are dedicated to routine care.

Administrators of a hospice decide the number of visits, and set up the schedule for each nurse. A day for a hospice nurse involves driving to several patients as determined by the administrator, giving them care, writing up reports for the hospice physician and administrator, and refilling supplies. From my interviews with hospice administrators and readings on the industry, hospice staff do not drive out from the hospice office to visit patients; rather, they start from their home and visit patients. Thus, the exact location of the hospice office does not directly influence driving time.

The care given by hospice nurses is relatively low-skill compared to other medical providers. Hospices are not responsible for curative treatment; they are tasked with pain management and ease-of-living for a dying patient. Examples of hospice care include administering pain medication, providing medical supplies like oxygen and bandages, dressing bedsores, giving physical and speech therapy, helping with bathing and feeding, temporarily substituting as the primary caregiver, tending to emotional and spiritual needs of the patient, and giving grief counselling to the family after the patient passes away.⁵

It is important to note that the hospice decides how many visits to make to each patient and what care is to be provided during each visit. As is explained below, the patient does not incur any additional cost for more visits, and so would plausibly like as many visits as possible.

2.2 Medicare reimbursement and quality competition

Hospices operate under a fixed-price setting. Most hospice patients are covered by Medicare or Medi-Cal (verified from my data), which reimburses hospices a fixed rate for every day the patient is enrolled (i.e. a per-diem rate). The fixed rate is set exogenously and does not depend on the number of visits that the hospice staff makes. For a patient covered by Medicare or Medi-Cal, hospice care is essentially free.

⁵Hospice staff do not function as the primary caregiver of the patient - someone in the patient's residence (a family member if the patient resides at home, or a nurse if the patient resides in a nursing facility) has to be the primary caregiver. But hospice nurses assist the primary caregiver, and sometimes may substitute in for a period of time to give respite to the primary caregiver.

Hospices primarily compete in quality, rather than price or location. Higher quality is more costly to provide, and hospices are not directly reimbursed for higher quality, so the main economic motive for providing higher quality is to attract more patients.

I claim that the frequency with which a hospice visits its patients (visits-per-patient-day) reflects the quality and effort the hospice provides. A hospice which makes more visits is constantly checking and adjusting to the patient's condition, can manage symptoms better, and is likely to arrive quickly in case of an emergency. This measure is thoroughly discussed and validated in [Alam \(2025\)](#). I show empirically that markets facing greater competition see more frequent visits to patients, suggesting quality improvements.

Since more visits are more costly while the per-patient reimbursement is the same, greater competition leads to higher quality choice and smaller per-patient profit margins.

2.3 Data

The main dataset comes from Home Health Agencies And Hospice Annual Utilization Reports compiled by California Department of Health Care Access and Information (HCAI). Hospices in California are required to submit yearly utilization reports to HCAI where they report firm-level statistics for the year. This includes measures such as current location, total number of patients, total visits per year (broken down by type of staff), hospice characteristics, aggregate characteristics of patient pool, cost breakdowns, etc. I collect and clean utilization reports of hospices for 2002-2018. I complement this with additional data sources on population sizes (by age), mortality rates, and Medicare hospice reimbursement rates for California over the same time period.

A market is defined to be at the level of a county.⁶ Market size is constructed by multiplying

⁶While some hospices offer services to multiple counties, I can rule out clusters of counties as a market definition using data. Specifically, the dataset contains a breakdown of a hospice's patient pool by patients' counties of residence. Most of a hospice's patients originate from the county where the hospice is located. The only exception are Yuba and Sutter counties which have a large overlap of patients, and so I combine them into a single market. The number of county-years enumerated later reflect this alteration.

the population size of each age for a market with the corresponding mortality rate⁷; that is, market size is the expected number of deaths happening in a market in a year.

For this project, I drop Los Angeles county (LA) from my main analysis due to recent allegations of fraud in hospices located in LA. The chief complaints are that individuals may be opening up fake hospices that provide no service and bill Medicare for fraudulent claims. According to the state auditor’s report, the problem appears to be localized in LA. The worry with including LA data in my analysis is that if there is widespread fraud in LA hospices, then their strategic decisions (and data reporting) may be more problematic. See Section J for discussion and links to reports. For completeness, I present results including LA in the Appendix.

After data cleaning, I have a panel dataset of 45 counties in California, with 700 county-years. The remaining counties do not observe hospice entry over my dataset’s time period.

Since the cost data is critical to my paper, I now present additional details about these variables. The cost breakdown includes costs incurred for administrative and general services, cost of program supervision, cost of visiting services, costs for specific hospice services (such as drugs, transportation, etc.), and various other costs. See Figure 19 for the full list of cost variables.

The cost data are exactly the ones that hospices report for Medicare Cost Reports; in fact, the HCAI forms specifically ask hospices to copy over the data from their Medicare Cost Reports. Thus, the cost data is consistent with what hospices report to Medicare.

The reported costs also do not affect the reimbursement received by hospices from Medicare. First, Medicare reimbursement depends on days of patient enrollment, not individual hospices’ production costs. Second, while Medicare rates for a county are adjusted by a “wage index”, this index is calculated at the CBSA level using external data on wages, not hospice costs.⁸

The accuracy of this accounting data is further secured by two institutional guardrails. First, Medicare Hospice Cost Reports are subject to serious penalties for falsification and to audits by

⁷These data are publicly available online; see Appendix C. For population, the data lists the population for each age in California for every year from 2000 to 2020. The death rates are available at the US level, for each year, and for male and female. I average over the male and female deaths to obtain the death probability for each age by year.

⁸Link for calculation of wage index: <https://www.federalregister.gov/documents/2025/08/05/2025-14782/medicare-program-fy-2026-hospice-wage-index-and-payment-rate-update-and-hospice-quality-reporting>

federal contractors. Second, the data is subject to desk-audit before release by HCAI.

2.4 Cost measures

I define the following cost measures:

1. “Total visit cost” should include the cost components in the cost report that stem directly from hospice visits. I measure it as the sum of the following cost report sections: “Visiting Services Costs”, “Hospice Service Costs” (except costs of radiation therapy and chemotherapy), and “Other Program Costs”.⁹
2. “Operating cost” should include the cost components in the cost report that relate to the fixed cost of operating. I measure it as the sum of the following cost report sections: “General service costs - administrative and general”, “Other hospice service costs”, “Program supervision costs – hospice program/team supervision (non-visit wages)”, and “Volunteer program costs”.

The per-visit cost for hospice j in year t is estimated as:

$$\text{Per-Visit Cost}_{jt} = \frac{\text{Total Visit Cost}_{jt}}{\text{Total Visits}_{jt}}$$

⁹The cost report further defines Visiting Services Cost as comprising the following cost components: “Physician Services”, “Nursing Care”, “Rehabilitation Services (PT, OT, Speech)”, “Medical Social Services – Direct”, “Spiritual Counseling”, “Dietary Counseling”, “Counseling – Other”, “Home Health Aides and Homemakers”, and “Other Visiting Services”.

The cost report further defines Hospice Service Cost as comprising the following cost components: “Drugs, Biologicals, and Infusion”, “Durable Medical Equipment / Oxygen”, “Patient Transportation”, “Imaging, Lab, and Diagnostics”, “Medical Supplies”, “Outpatient Services (including ER Dept.)”, “Radiation Therapy”, “Chemotherapy”, and “Other Hospice Service Costs”. Costs of radiation therapy and chemotherapy are dropped from the analysis as these are unlikely to be a part of a hospice’s visiting cost or usual operating cost, since hospices are not responsible for curative treatment. The component “other hospice service costs” are included under operating cost as it is unclear what this includes.

I exclude costs of inpatient care, fundraising, and bereavement care from the main analysis. Very few hospices have inpatient units or serve inpatient care days. Costs of bereavement care is not part of the core hospice service. For robustness, I include these under total visit cost and get qualitatively same results. Fundraising is left out as it is not part of hospice care.

See Figure 19 for the full layout of the cost report.

For robustness, I redefine “total visit cost” as including more and fewer cost components and rerun all my empirical analysis. The results are all qualitatively the same.

where "total visits" is the total number of physical visits made by all hospice staff.

This is an empirical measure of the average cost of making a visit to a patient. Since the numerator includes total visit cost, this means that the fixed costs of operation ("operating cost") are excluded when calculating this average.

3 Descriptive Evidence

In this section, I present key descriptive statistics from the dataset. First, I show distributions of key variables such as per-visit costs and competition levels. Second, I show how the number of hospices have grown tremendously between 2002-2018, and that this growth is mostly driven by entry of for-profit hospices. Third, for-profits have improved patient access by serving more patients than non-profits, while non-profits have improved access by operating in small markets eschewed by for-profits. Fourth, I show that per-visit costs are highly persistent, which when coupled with the importance of visit costs in overall hospice costs, suggests that it may play an important role in market dynamics. Fifth, I show that per-visit costs are lower in counties with more hospices, which is the key pattern this paper aims to examine.

3.1 Summary statistics

Table 1 shows summary statistics of some of the key variables in my dataset. Notably, per-visit costs are around \$100 with significant dispersion across hospices. Total cost incurred from visits forms a significant share of a hospice's overall costs, with a median of 54%. There is little concentration in large markets, with a median HHI of 735.4.

To get a sense of the level of competition in the hospice industry, Table 2 shows the distribution of hospices across counties and years. A few large counties account for the majority of hospices in California. The number of hospices in these counties has been increasing over time.

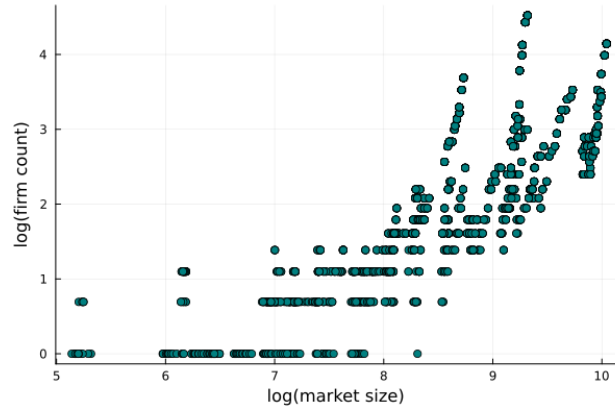
I find that the number of hospices in a market is strongly positively correlated with the expected number of deaths in that market; see Figure 1 for a graphical illustration, and Table 16 for regres-

Table 1: Summary statistics

Variable	10%	25%	50%	75%	90%
Per-visit cost	63.58	81.47	108.78	144.46	186.24
Total visit cost / Total cost	0.36	0.47	0.57	0.66	0.76
Visits per patient-day	0.54	0.65	0.8	1.01	1.29
Average length-of-stay	37.09	44.39	53.71	65.39	79.77
Patients	38.0	92.0	246.0	488.0	832.4
Market share (%)	0.55	1.82	6.69	23.62	58.29
HHI	597.41	857.77	1604.35	3053.62	6169.71

Notes: For this table, “market share” denotes the within-hospice market share, i.e., the denominator is the total number of observed hospice patients, not the number of potential patients. HHI is calculated using within-hospice market shares.

Figure 1: Scatterplot of market size and number of hospices



Notes: Each point represents a county-year observation. Market size is measured as the expected number of deaths in that county-year.

sion results. Thus, I will use the expected number of deaths in a market as an instrument for level of competition in regressions.

3.2 Market dynamics, For-profits, and Access to Hospice Care

To visualize market dynamics in the hospice industry, Figure 2 shows the evolution of entries, exits, and total hospices over time. The key takeaway is that the number of hospices in California has been increasing rapidly over time, and is mostly driven by entry of for-profit hospices. Number of nonprofits has remained roughly the same since 2007. The total entries and exits in my dataset,

Table 2: Distribution of hospices across counties and years

County	2002	2009	2013	2017
San Bernardino	6	20	28	84
Orange	11	18	27	54
Ventura	5	17	23	34
Riverside	9	14	23	31
San Diego	11	15	19	29
Alameda	7	9	11	20
Sacramento	5	7	9	18
Santa Clara	6	7	12	14
Contra Costa	4	6	9	12
Fresno	3	4	6	10
San Mateo	3	4	5	9
Kern	3	8	8	8
San Joaquin	2	2	4	7
Santa Barbara	2	3	5	6
Sonoma	4	5	6	6
Placer	2	4	5	5
San Francisco	6	5	6	5
Shasta	1	2	2	4
Solano	2	2	3	4
Stanislaus	2	3	3	4
Butte	4	3	3	3
Imperial	1	1	1	3
Monterey	3	2	3	3
San Luis Obispo	1	2	3	3
Tulare	1	2	3	3
Sutter	2	2	2	3
El Dorado	2	2	2	2
Marin	1	1	2	2
Merced	2	3	2	2
Nevada	2	2	2	2
Amador	1	1	1	1
Humboldt	1	1	1	1
Kings	1	1	1	1
Mariposa	1	2	1	1
Mendocino	1	1	1	1
Napa	1	1	1	1
Santa Cruz	1	1	2	1
Siskiyou	2	3	3	1
Yolo	1	1	1	1
Lake	0	1	1	1
Tehama	0	1	1	1
Tuolumne	1	0	1	1
Inyo	0	0	0	1

Notes: Yuba and Sutter counties have a large overlap of patients, and so I combine them into a single market.

broken down by for-profit status, are shown in Table 3, and corroborate the broad patterns from the graphs.

How has this growth affected patient access to hospice care? I argue that both for-profits and non-profits have improved patient access but in different ways.

1. For-profits: Over time, the share of patients treated by for-profits have grown tremendously,

and now covers significantly more hospice patients than non-profits (Table 2).¹⁰

2. Non-profits: Non-profits have improved access by operating in smaller markets eschewed by for-profits. Table 4 contrasts the market size of i) counties with non-profits but without for-profits, and ii) counties with for-profits. The former has lower market size than the latter, suggesting that non-profits are willing to operate in smaller markets that for-profits may not find profitable enough, and therefore plays an important role in underserved areas. This is partly aided by donations; Table 5 shows that non-profits receive more donations-per-patient in counties with fewer hospices, suggesting that donations help non-profits operate in smaller markets.

Interestingly, 2006-2007 saw a large number of hospices (48 in total) switch from non-profit to for-profit status. This transition explains a large part of the patient share between the two types. These 48 hospices are spread out across 17 counties, and some seem to be part of larger chains transitioning to for-profit status. In total my data sees 65 transitions from non-profit to for-profit status, and 16 transitions from for-profit to non-profit status.

3.3 Market size

Recall that market size is measured as the expected number of deaths in a market. Appendix L.2 shows the distribution of expected deaths across market-years, as well as the evolution of expected deaths for a subset of counties. I also show the same figures for $\log(\text{expected deaths})$ and for residualized expected deaths (residualized on county fixed effects). Market size shows heterogeneous trends across counties, ranging from steady increases to stability or declines.

¹⁰This is partly through drawing in new patients who otherwise would not have chosen hospice care (i.e. the “outside option”), rather than entirely capturing patients previously going to non-profits. Prior research by Alam (2025) finds market expansion effects of hospice entry to be significant. Following Chung and Sorensen (2018), patient access should be tied to market expansion effects, i.e. does a new hospice entirely steal patients from other hospices, or does it bring in new patients who would not have otherwise enrolled in hospice care. This can be studied using the demand estimation method of Berry (1994) and checking if the nested logit parameter is significantly below 1, a strategy outlined by Chung and Sorensen (2018). Alam (2025) estimates a detailed demand model for hospices in California and indeed finds market expansion effect.

Figure 2: Time series plots of market dynamics.

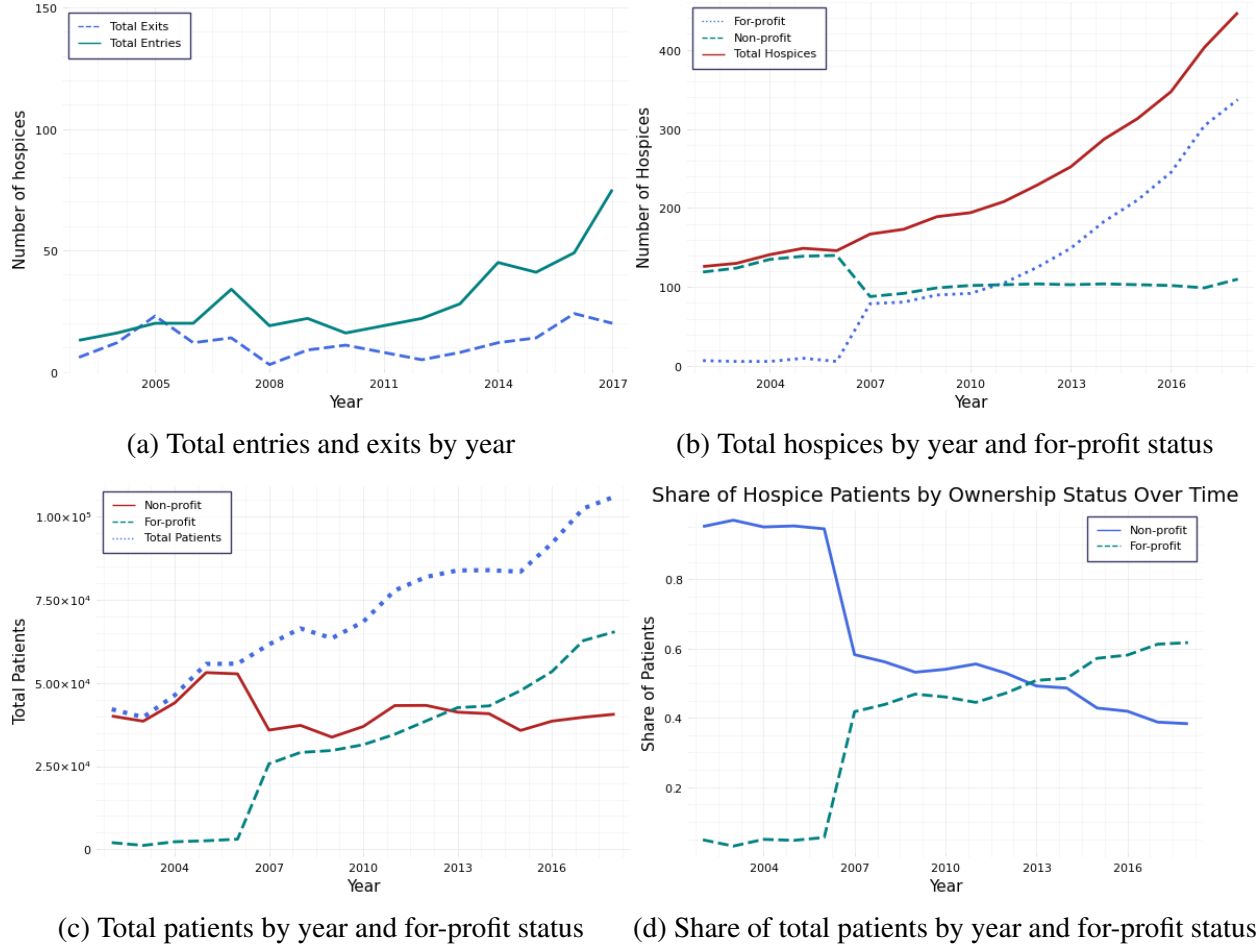


Table 3: Total entries, exits, and hospices.

Variable	Value
Total Exits	177
Total Entries	498
Total Unique Hospices	624

(a) Total entries, exits, and hospices.

For-profit Status	Total Entries	Total Exits
0	127	87
1	371	90

(b) Entries and Exits by for-profit status

Table 4: Market size by presence of for-profits.

Variable	Counties without For-profits	Counties with For-profits
Mean market size	2229.83	7067.76
Median market size	1207.7	5298.55

Notes: “Counties without For-profits” denotes counties which have non-profits but no for-profits. “Counties with For-profits” denotes counties which have for-profits (and may or may not have non-profits). Market size is measured in expected number of deaths per year.

Table 5: Regression of donations per patient on competition level.

	Donations per patient
log(firm count)	-88.411*** (10.727)
year Fixed Effects	Yes
Obs.	1,414

Notes: Robust standard errors in parentheses.

3.4 Hospice costs and visits

Next, I examine the persistence of per-visit costs within a firm. I find that per-visit costs are highly persistent within a firm over time, with an autocorrelation coefficient of 0.95 (Table 6).

This raises an important point. The high level of persistence means that current per-visit costs are a strong predictor of future per-visit costs. Since visit costs form a significant share of total costs (with a median of 54% from Table 1), this means that current per-visit costs have strong implications about future costs and profits. This makes it highly likely that per-visit costs affect entry and exit decisions of firms, which I corroborate in my empirical analysis. This analysis is similar to [Collard-Wexler \(2011\)](#), who studies persistence of productivity in the ready-mix concrete industry, and argues that greater persistence of productivity means current productivity is a good indicator of future profits.

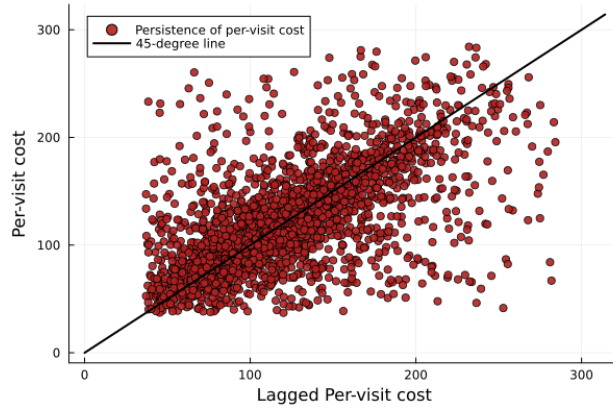
Hospices in more competitive markets make more visits-per-patient-day, as shown in Figure 16 and Table 21. Hospices which make more visits-per-patient-day also have greater market share – see Table 31 for a descriptive regression of patient volume on visits-per-patient-day, controlling

Table 6: Regression of current per-visit cost on lagged per-visit cost.

	Per-visit cost
Lagged per-visit cost	0.945*** (0.007)
Obs.	2,712
R-squared	0.902

Notes: No intercept was included in the regression. Standard errors clustered at the county-level in parentheses.

Figure 3: Plot of current versus lagged per-visit costs.



for other factors. In more competitive areas, market share reallocates towards higher visits-per-patient-day hospices (Figure 17 and Table 22). These patterns are explored in more detail for a smaller set of counties in Alam (2025).

To motivate the research question of this paper, I plot various averages of per-visit costs (Figure 4). First, I plot county-level-mean of per-visit costs against county-level firm numbers for 2016 and 2017. Strikingly, I find that per-visit costs are lower in counties with more hospices. Second, I plot the year-level-mean of per-visit costs, and find that it is declining rapidly over time (Figure 5).

4 Motivating evidence

In this section I dig into the relationship between per-visit costs and competition. First, I regress per-visit costs on competition level to show that a negative relationship exists, and is robust to

Figure 4: Variation in mean per-visit costs in the cross-section.

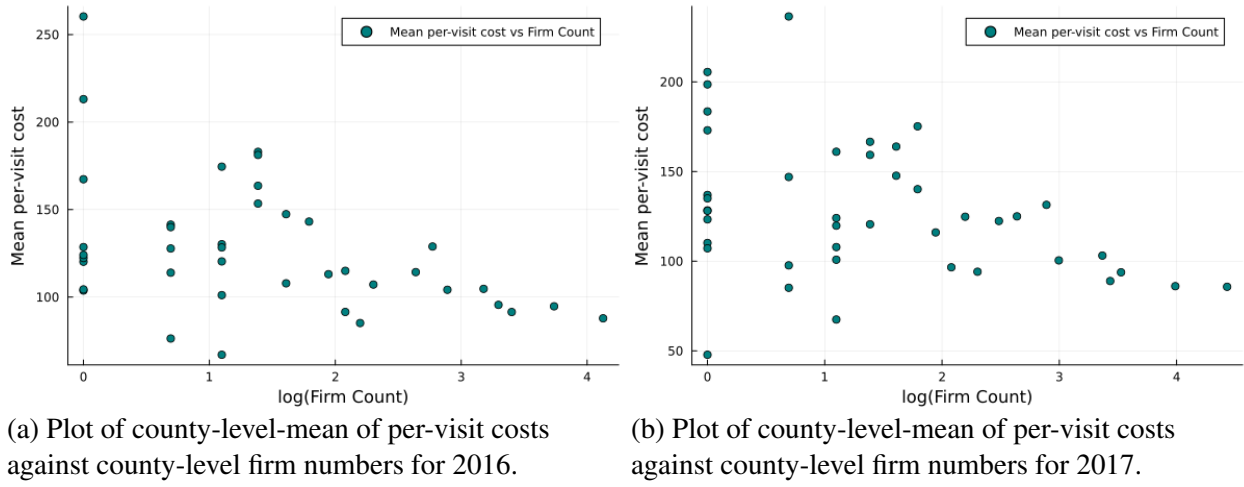
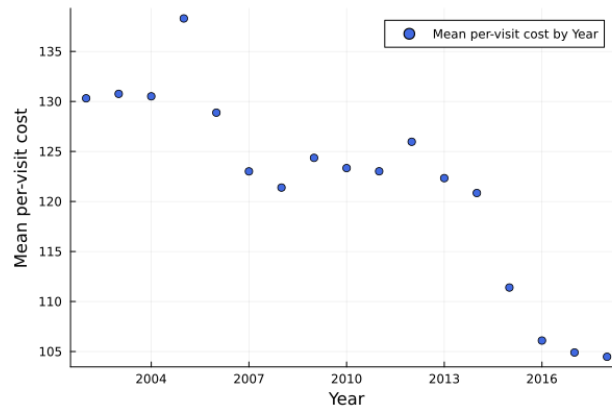


Figure 5: Plot of year-level-mean of per-visit costs over time.



confounding factors. Second, I use a grouped-quantile IV regression to show that the distribution of per-visit costs shifts left with greater competition. Third, I hypothesize that the negative relationship may be driven by several economic mechanisms that subsequent sections will examine. Fourth, I present descriptive evidence about cost differences between for-profits versus non-profits, entrants versus incumbents, and exiting versus surviving hospices.

I am interested in the link between competition and per-visit costs in the hospice industry. In my setting, greater competition can plausibly lead to lower per-visit cost in a market. Under Medicare's fixed price setting, greater competition forces hospices to raise visit frequency to compete on quality, squeezing profits. Low-cost providers are better able to sustain profit margins, so greater

competition selects for — and pressures incumbents toward — lower per-visit costs. I discuss this in detail in Section 4.3. Figure 4 suggests this may be the case, and below I investigate the robustness of this pattern with regression methods.

4.1 Regressions of per-visit costs

To see the link between per-visit costs and level of competition in my data, I regress per-visit cost on number of firms. While doing so, I attempt to control for possible confounding factors:

1. Firms may be more likely to enter markets where costs are exogenously declining. I tackle this in several ways.

First, I use market size (i.e. expected number of deaths in a market) as an instrument for competition level. This is motivated by the idea that market size is a demand component that shifts the number of firms, and is orthogonal to cost shocks. I argue that this is an exogenous demand shifter that instruments for long-run market structure and affects firms’ dynamic decisions.¹¹ More expected deaths in a market means more potential patients, which will lead to greater hospice entry. However, greater expected deaths shouldn’t directly influence per-visit cost of hospices. See Table 16 for the first-stage of this instrumental variable regression. See Section 6.1 for further discussion on the validity of the instrument.

Second, I include polynomials of lags of market-level-mean per-visit costs as controls. Lags of market-level-mean per-visit costs help predict the trend of costs at the market level. The “polynomial of lags” functional form is motivated by the idea that firms and entrants are forming expectations about future costs based on past trends (which then influence dynamic decisions), so to capture such expectation formation without imposing any behavioral assumption, the function of lagged costs should be very general.

Third, I include county fixed effects in my regressions to control for time-invariant factors

¹¹This is similar to Backus (2020), who uses demand shifters such as total permits issued to instrument for competition levels of concrete plants.

I also run the regressions without the instrument and obtain similar results.

affecting per-visit costs such as county area.

2. I include operating cost as a control to account for additional factors of production that may affect the marginal cost of a visit.
3. Cost functions may depend on quantity produced (reflecting economies or diseconomies of scale). Thus, as robustness checks, I include second-order polynomials of quantity in the regressions. I measure quantity via total patient days and total visits by hospices.¹²

Thus, the regression equation is:

$$\log(c_{jt}) = \beta_m + \beta_f f_{m(j)t} + X'_{jt} \beta_x + \varepsilon_{jt} \quad (1)$$

where β_m are market fixed effects, $f_{m(j)t}$ is competition level (instrumented with expected number of deaths), X_{jt} are operating cost, second-order polynomials of one-year-lags and two-year-lags of market-level-mean per-visit costs, and other covariates.

Table 7 shows the results from 2SLS regression of per-visit costs on number of firms. The results suggest that markets with more firms have hospices with lower per-visit costs. See Table 16 for the first-stage of this instrumental variable regression. I also perform robustness checks by measuring competition level with number-equivalent-HHI and find similar results (Section K.1).

¹²Given the industry details, economies of scale is more likely. Recall that hospice nurses usually start from their own home and drive out to multiple patient residences over the course of the day. If a hospice has more patients, that is more likely to reduce distance travelled to each patient (as subsequent visits are probably closer), resulting in economies of scale. [Bensaid et al. \(2022\)](#) finds evidence of economies of scale in the home health industry in France. In my estimation, I find some evidence of economies of scale: when I measure volume with total visits, I find per-visit costs to be somewhat declining in total visits.

Note that even if economies of scale were present but I did not control for it in my regressions, it would bias my coefficients in the opposite direction to what I find. If there are economies of scale, then greater competition (more firms) would lead to lower quantity per firm, which would increase per-visit costs. Thus, if I do not control for economies of scale, it would bias my coefficient on competition upwards (i.e. towards zero). Since I find a negative coefficient, this means that the true effect would be even more negative than what I estimate.

An incorrect argument for number of hospices driving down per-visit costs is that more hospices mean greater concentration of hospice offices within a market, which reduces travel time between hospice office to patients. This is unlikely to be true. First, hospices do not drive out from their office to visit patients; rather, they start from their home and visit patients. Thus, the exact location of the hospice office does not directly influence driving time. Second, it would imply that costs of all hospices decline as more hospices enter. In my empirical analysis, I find that cost of entrants is lower than cost of incumbents.

Table 7: Regression of per-visit costs on competition and controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	-0.125*** (0.020)	-0.123*** (0.020)	-0.129*** (0.022)	-0.104*** (0.018)
Total patient days / 1000		0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.038** (0.011)	
Total Visits / 10,000 squared			0.000 (0.000)	
log(operating cost)	-0.008 (0.008)	-0.013 (0.010)	0.031* (0.013)	-0.014 (0.008)
Inpatient unit services				-0.032 (0.045)
Pediatric program services				0.018 (0.051)
Bereavement services for non-hospice survivors				0.090*** (0.025)
Adult day care services				0.267* (0.104)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: IV regression of log(per-visit cost) on log(firm count) with county fixed effects and controls. Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

4.2 Quantile analysis

While the 2SLS regression shows how the average per-visit cost shifts with competition, I drill further into the data to examine changes in per-visit cost distributions as a result of competition. First, I illustrate the distribution of per-visit costs in three counties. Second, I run grouped-quantile regressions to rigorously show how the distribution of per-visit costs changes with competition. I find that increase in hospice competition shifts the distribution of per-visit costs left.

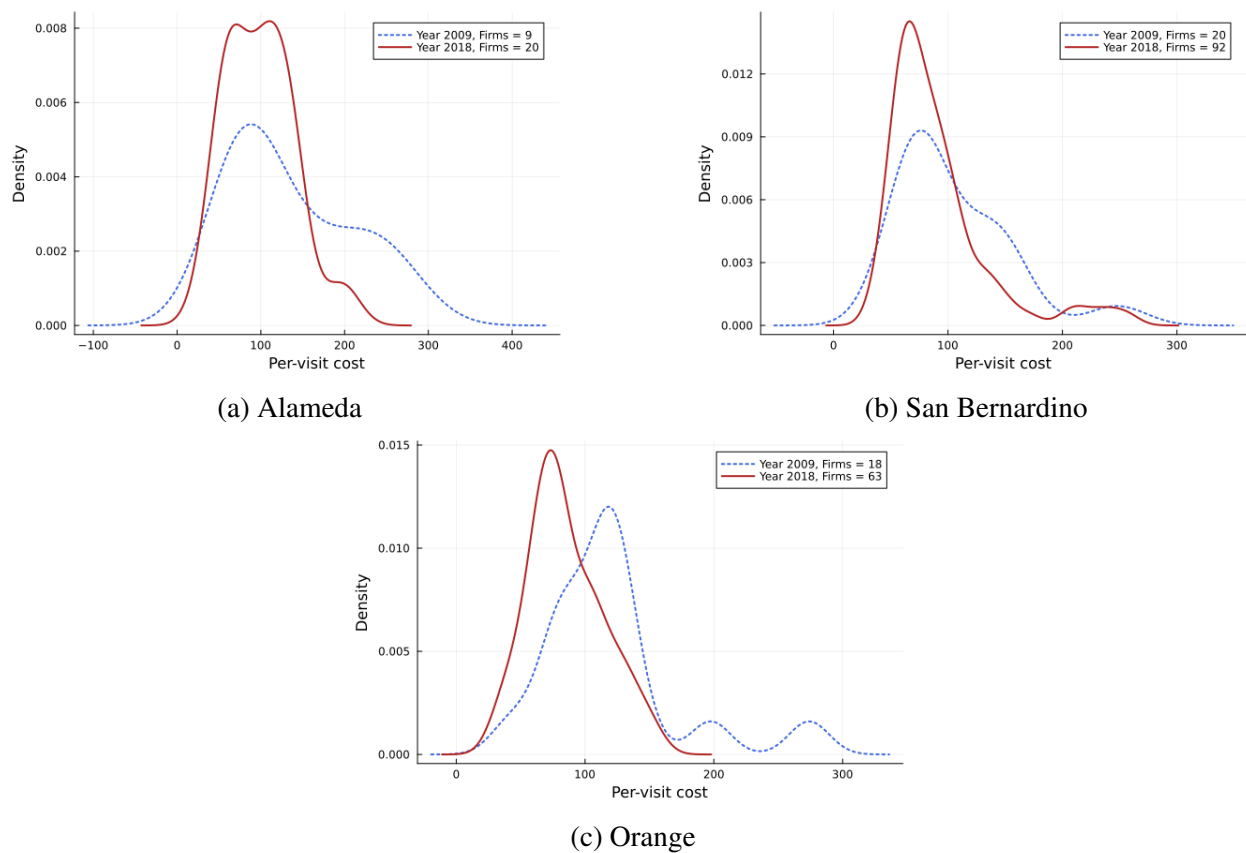


Figure 6: Smoothed densities of per-visit cost in multiple counties.

For a simple demonstration, Figure 6 plots smoothed densities of per-visit costs in years 2009 and 2018, during which time the level of competition increased substantially. It does so separately for three counties. This demonstrates how the distribution of costs change as level of competition rises. All 3 graphs show a similar pattern – as level of competition increases, the distribution of per-visit costs shifts left.

Table 8: Grouped-quantile regressions of per-visit costs on competition.

Quantile	Coefficient	S.E.	Quantile	Coefficient	S.E.
0.1	-62.0691	47.1972	0.1	-93.569	59.2587
0.2	-57.7973	46.5341	0.2	-81.8432	59.0394
0.3	-54.821	46.1504	0.3	-72.2143	59.2026
0.4	-46.7867	45.9773	0.4	-57.9146	59.4547
0.5	-39.649	46.1329	0.5	-47.2737	60.1173
0.6	-30.7219	45.9584	0.6	-30.1996	59.828
0.7	-22.468	46.4914	0.7	-10.7681	60.5942
0.8	-23.2499	47.4024	0.8	-3.16611	62.0975
0.9	-23.9292	49.1447	0.9	3.44088	64.5013
(a)			(b)		

Notes: Grouped-quantile IV regression of per-visit costs on competition, county fixed effects, and semi-parametric order-statistic bias correction. Table 8b also includes second-order polynomials of one-year-lag and two-year-lag of county-level-mean of per-visit costs. Robust standard errors are reported under S.E. See Section A.1 for details about implementation.

Next I run grouped-quantile regressions to measure how the distribution of per-visit costs changes with competition within my dataset. This follows the methods of Backus (2020), Chetverikov et al. (2016), and Larsen et al. (2023); see Section A.1 for details about implementation. I regress market-level quantiles on competition level in the market, using market size as an instrument for number of firms. Following Backus (2020), I use third-degree polynomials of the number of firms to remove an order-statistic bias. I include county fixed effects in all regressions, and every parameter is estimated separately for each quantile.

Let k denote the k th decile, and $\rho_{mt}^{(k)}$ be the k th decile of the market-level distribution of per-visit costs. The grouped-quantile regression equation is:

$$\rho_{mt}^{(k)} = \alpha_m^{(k)} + \alpha_f^{(k)} f_{mt} + X_{mt}' \alpha_x^{(k)} + g^{(k)}(n_{mt}) + v_{mt} \quad (2)$$

where f_{mt} is the level of competition and n_{mt} is the number of firms in the market. The semiparametric correction for the order statistic bias, $g^{(k)}(\cdot)$, is a third-degree polynomial of the number of firms in the market, and is estimated separately for each k . As a robustness check, I include

market-level controls X_{mt} by having lags of market-level-mean per-visit costs. In Tables 8a and 8b, f_{mt} is log of number of firms.

The results are shown in Tables 8a and 8b (which additionally includes second-order polynomials of one- and two-year lags of market-level-mean per-visit costs). Most quantiles shift left with more firms in the market, with the results being stronger at lower quantiles than higher quantiles.

4.3 Why does competition lower per-visit costs?

Why does competition lower per-visit costs? Under Medicare’s current reimbursement scheme, a hospice’s per-patient profit equals the per-patient reimbursement minus its total visit cost, defined as the product of its per-visit cost and visits-per-patient. An increase in competition forces firms to compete on quality, thus raising visits-per-patient to secure market share. This reduces per-patient profits as reimbursement rates are fixed. Providers with lower per-visit costs maintain higher profit margins. Thus, only low-cost firms can profitably enter or survive in highly competitive markets, and incumbents face pressure to reduce their own per-visit costs. This means that the distribution of per-visit costs is likely to shift left with greater competition.¹³ See Section 6.5 for further discussion about economic mechanisms and connecting them to the empirical analysis in the rest of the paper.

Therefore, I argue that the coefficient on competition in Table 7 could comprise three mechanisms:

1. Entry-composition effect: Greater competition occurs through entry of lower-cost firms, i.e. the composition of firms in the market shifts towards lower-cost firms.
2. Exit-selection effect: Greater competition forces higher-cost firms to exit the market.
3. Incumbent-treatment effect: Greater competition forces incumbents to lower their per-visit costs. This may be due to reduction in X-inefficiency or managerial slack; see [Leibenstein \(1966\)](#), [Perelman \(2011\)](#), and [Backus \(2020\)](#).

¹³This claim about per-visit cost measure should not be confused with the total cost per patient. The total cost of serving a patient may rise overall if greater quality competition forces all hospices to increase their visits-per-patient.

Table 9: Regression of per-visit costs on for-profit indicator and county-year fixed effects.

	log(per-visit cost)
For-profit	-0.219*** (0.050)
County $\times$ Year Fixed Effects	Yes
Obs.	3,066

Notes: Standard errors clustered at county-level in parentheses.

Each of these three mechanisms can lead to the coefficient on competition level being negative. Which of these mechanisms are present in the California hospice industry? The rest of the paper sheds light on this question.

4.4 Motivating evidence about mechanisms

I now present descriptive evidence about cost differences between for-profits versus non-profits, entrants versus incumbents, and exiting versus surviving hospices. This will also help clarify the variation in data that the next section’s regression methods will use.

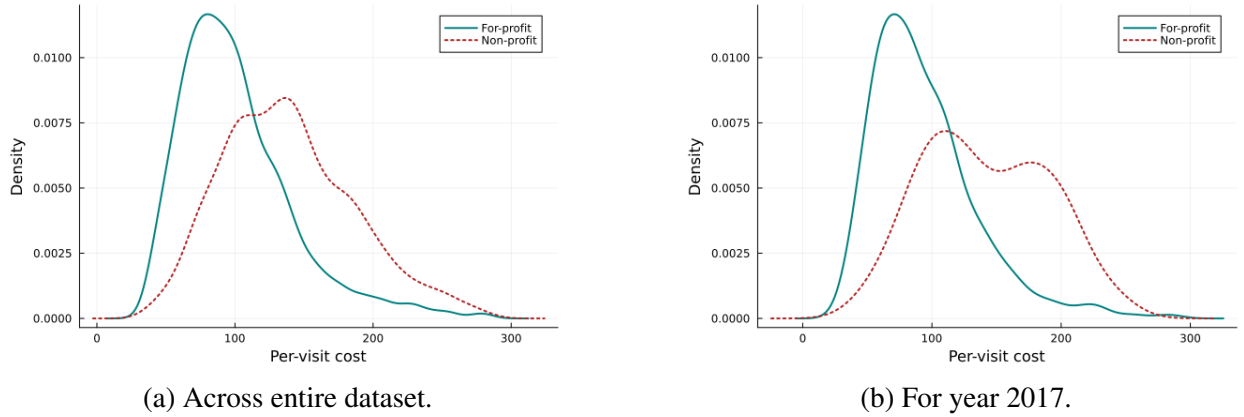
4.4.1 Cost differences between for-profits and non-profits

Since the growth in number of hospices is mostly driven by for-profit entry, I study whether for-profit hospices have lower per-visit costs than non-profits. I regress per-visit costs on for-profit status and county-year fixed effects. I also plot smoothed densities of per-visit costs broken down by for-profit status.

I find strong evidence that for-profit hospices have lower per-visit costs than non-profit hospices. See Table 9 for regression results, and Figure 7 for distribution of per-visit costs by for-profit status. The density plots show that for-profit hospices have a cost distribution that is shifted left relative to non-profits, indicating lower costs. I also plot the density for year 2017 separately alone to rule out possible across-year variation as a confounder and find the same result.

In Section 5, I will show that this conclusion holds on a set of comprehensive regressions over

Figure 7: Density of per-visit costs by for-profit status.



the full dataset.

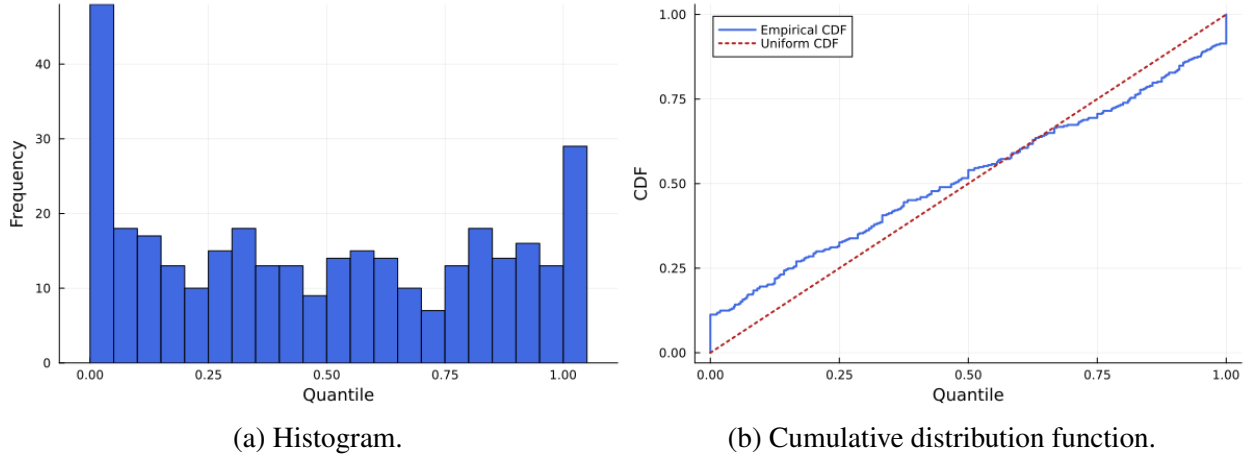
When combined with the fact that for-profits are driving most of the entry into the California hospice industry, this suggests that for-profit entry is an important driver of the entry-composition effect.

4.4.2 Per-visit costs of entrants

For clarity, I first graphically study the costs of hospice entrants relative to incumbents. For each entrant, I compute the quantile of its per-visit cost in the empirical cost distribution of other hospices in the same county-year (excluding itself). This “leave-one-out” approach ensures that each hospice’s quantile reflects its position among rivals and not include its own value. If entrants have low quantile values, it means they enter with costs lower than most incumbents; high quantile values indicate entrants with relatively high costs.

I illustrate my findings with the following graphs. First, I plot a histogram showing the frequency with which entrants fall into different quantiles of the rivals’ cost distribution in the county-year. Second, I plot the cumulative distribution of these quantiles. This empirical CDF is compared to the CDF of a uniform distribution (the 45-degree line, which represents the benchmark of entrants randomly drawn from the county-year cost distribution). Together, these plots shed light on whether entrants are more efficient than incumbents: if the plots are skewed towards lower

Figure 8: Quantiles of per-visit costs of entrants in a county-year, using leave-one-out.



Notes: These plots limit to markets with greater than 5 hospices for reliable quantile calculation.

quantiles, then entrants tend to have lower costs than their local competitors. If the plots are more uniform, entrants are spread evenly across the cost spectrum. Figure 8 shows the results, limiting to markets with greater than 5 hospices for reliable quantile calculation; and Figure 15 limits to markets with greater than 10 hospices for robustness.

Both plots show that entrants tend to have lower per-visit costs than incumbents, as they are more likely to appear at lower quantiles. More specifically, this suggests entrants have lower per-visit costs than their rivals in the same county-year. Most striking is that entrants often have lower costs than all incumbents currently in the market, as illustrated by the histogram.

Next, I show suggestive evidence that later entrants have lower costs than earlier entrants. For each entrant, I calculate their mean per-visit cost over all years in my dataset, which I call their “mean lifetime per-visit cost”. I regress the mean lifetime per-visit cost of a hospice on its entry time and other controls (Table 10). For the regression, I only include hospices entered after 2002 and those which never changed their for-profit status over my dataset. I measure entry time as the number of years since 2002 for Table 10. I find that later entrants have lower mean-lifetime per-visit costs than earlier entrants. To aid visualization, I plot the mean lifetime per-visit cost of hospices against their entry time in Figure 18, which also shows a downward trend.

In Section 5, I will show that this conclusion holds on a set of comprehensive regressions over

the full dataset.

Overall, this suggests that entrants being lower-cost than incumbents and prior entrants is an important driver of the entry-composition effect.

Table 10: Regression of mean lifetime per-visit cost on entry time and for-profit indicator.

	Mean lifetime per-visit cost	log(mean lifetime per-visit cost)
	(1)	(2)
Entry time	-1.204* (0.581)	-0.016** (0.006)
For-profit	-15.404* (6.012)	-0.137** (0.044)
County Fixed Effects	Yes	Yes
Obs.	425	425

Notes: Standard errors clustered at the county level in parentheses. Entry time is measured as number of years since 2002. For each entrant, I calculate their mean per-visit cost over all years in my dataset, which I call their “mean lifetime per-visit cost”.

4.4.3 Per-visit costs of exiting hospices

In this section I study if higher-cost hospices are more likely to exit. Since exiting the market may require the hospice to incur additional costs such as severance payments, for this section I calculate “per-visit costs” as total visiting services costs divided by total visits. The intuition is that costs on visits shouldn’t vary based on whether the hospice will soon exit or not.

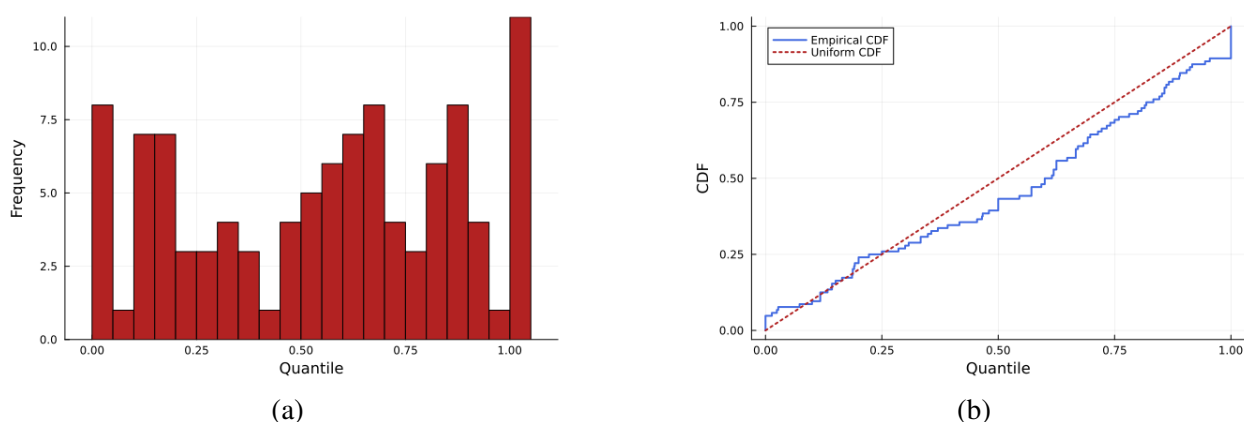
Similar to the analysis on hospice entrants, I first graphically study the costs of exiting hospices relative to its rivals. For each exiting hospice, I compute the quantile of its per-visit cost in the empirical cost distribution of other hospices in the same county-year (excluding itself). If an exiting hospice has low quantile values, it means they have costs lower than most incumbents, and vice versa.

I illustrate my findings with the following graphs. First, I plot a histogram showing the frequency with which exiting hospices fall into different quantiles of the rivals’ cost distribution in the county-year. Second, I plot the cumulative distribution of these quantiles. This empirical CDF

is compared to the CDF of a uniform distribution (the 45-degree line, which represents the benchmark of exits happening randomly across the county-year cost distribution). Together, these plots shed light on whether exiting hospices are less efficient than incumbents. If the plots are skewed towards higher quantiles, then exiting hospices tend to have higher costs than their rivals. Figure 9 shows the results, limiting to markets with greater than 5 hospices for reliable quantile calculation; and Figure 14 limits to markets with greater than 10 hospices for robustness.

The graphs suggest exiting hospices are slightly more likely to above-median in per-visit costs than their rivals. Most striking is that exiting firms often have higher costs than all incumbents currently in the market, as illustrated by the histogram.

Figure 9: Distribution of per-visit costs of exiting hospices



Notes: Figures 9a and 9b plot the quantiles of per-visit costs of exiting hospices in a county-year, using leave-one-out. These plots limit to markets with greater than 5 hospices for reliable quantile calculation.

To accurately quantify whether exiting firms do in fact have higher costs, I regress the binary variable of "Exit" on per-visit costs and county-by-year fixed effects (Table 11). The results show that higher per-visit costs are associated with a higher probability of exit. I also find that hospices with more patients are less likely to exit. (This is similar to Collard-Wexler (2011), who finds that firm size is negatively related to exit probability.)

Note that this regression does not say anything about the exit-selection effect of competition. The exit-selection effect says that competition forces higher-cost firms to exit, while the above regression says high cost firms are more likely to exit without controlling for competition. To

Table 11: Regression of exit indicator on patients and per-visit costs.

	Exit
log(patients)	-0.028*** (0.005)
log(per-visit cost)	0.033** (0.011)
County $\times$ Year Fixed Effects	Yes
Obs.	2,653
R-squared	0.196
Adj. R-squared	0.042

Notes: Standard errors clustered at county-level in parentheses.

study exit-selection effect directly, I regress exit on interaction of competition level and per-visit costs, along with other controls (Table 24). While the coefficient on the interaction term is not statistically significant – which seems to rule out exit-selection effect – the regression lacks power in general. I will return to the exit-selection effect in the next subsection, and confirm via a different method that in fact exit-selection is absent in my data.

5 Empirical Analysis

I examine the factors that drive the negative relationship between per-visit costs and competition using linear regression methods. First, I show that for-profits have lower costs than non-profits. Second, I show that entrants have lower costs than incumbents, and successive entrants have lower costs than prior entrants. Third, I show that greater competition does not lead to incumbents lowering their per-visit costs or for higher-cost incumbents to exit the market. Finally, I show that these results hold even after controlling for patient characteristics that may affect per-visit costs.

I now return to the three effects that can cause the coefficient on level of competition to be negative in Table 7: entry-composition effect, exit-selection effect, and incumbent-treatment effect.

The motivating evidence in the previous section suggests that including for-profit status and entry time measure in the regression for Table 7 can absorb the entry-composition effect. Thus,

after their inclusion, the coefficient on competition should be the sum of exit-selection effect and incumbent-treatment effect.¹⁴ Each of these mechanisms have a negative effect on per-visit costs. In this paper, instead of trying to disentangle these two effects, I focus on the combined effect.

I run a IV regression of per-visit costs on competition level, for-profit status, and entry time:

$$\log(c_{jt}) = \beta_m + \beta_f f_{m(j)t} + \beta_1 \text{For-profit} + \beta_2 \text{Entry time} + X'_{jt} \beta_x + \varepsilon_{jt} \quad (3)$$

I instrument competition level with market size (i.e. expected number of deaths in a market); see Table 16 for the first-stage of this instrumental variable regression. The results of this regression will tell us if for-profits have lower costs than non-profits, if later entrants have lower costs than earlier entrants, and if there is any remaining effect of competition on per-visit costs after controlling for these two factors.

The results are shown in Table 12. Confirming the patterns from the previous section, I find that for-profits have lower per-visit costs than non-profits, and later entrants have lower per-visit costs than earlier entrants.

Strikingly, once I include for-profit status and entry time measure, the coefficient on competition level is no longer statistically significant and is close to zero. This suggests that the negative relationship between per-visit costs and competition is not driven by the incumbent-treatment effect or exit-selection effect of competition, but rather by the entry-composition effect. In turn, this entry-composition effect is driven by i) for-profit hospices have lower per-visit costs than non-profit hospices, and ii) later entrants have lower per-visit costs than earlier entrants.

Since the exit-selection effect seems to be very weak given these results as well as Table 24, another way to check for the presence of incumbent-treatment effect is as follows. I restrict the sample to firms which entered by 2002, and run the IV regression of per-visit costs on number of firms, for-profit status, and other controls. Without the presence of exit-selection effect, the

¹⁴This is similar to the argument in Backus (2020). In that paper, the author runs a causal regression of productivity on competition, instrumenting it with demand shifters. He argues that the resulting negative effect is driven by the sum of “selection effect” (lower productivity firms forced out by competition) and “treatment effect” (firms improving their productivity in response to competition). Unlike my paper, Backus (2020) does not address entry by new firms.

Table 12: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.046 (0.033)	0.046 (0.033)	0.052 (0.038)	0.045 (0.032)
For-profit	-0.182*** (0.046)	-0.184*** (0.046)	-0.183*** (0.042)	-0.183*** (0.047)
Entry time	-0.016*** (0.004)	-0.016*** (0.004)	-0.018*** (0.003)	-0.015*** (0.004)
Entry before 2002	-0.028 (0.039)	-0.025 (0.039)	-0.025 (0.040)	-0.037 (0.040)
Total patient days / 1000		-0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.043*** (0.011)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.032** (0.010)	-0.024* (0.012)	0.010 (0.013)	-0.033*** (0.009)
Inpatient unit services				-0.058 (0.038)
Pediatric program services				-0.024 (0.047)
Bereavement services				0.057** (0.021)
Adult day care services				0.261** (0.095)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

coefficient on competition level should be only the incumbent-treatment effect.¹⁵

The results are shown in Table 18; once again, the coefficient on competition is not statistically significant and is close to zero.

Prior literature on palliative care (see Section I) has suggested that patient characteristics affect hospice costs. This raises the possibility that my results are driven by patient characteristics, if for-profits and later entrants are somehow primarily serving patients with characteristics that lead to lower per-visit costs.

To ensure that my results are not driven by patient characteristics, I rerun the previous regression while controlling for the following characteristics of a hospice's patient pool:

1. Share of patients with cancer diagnosis: Some papers in the literature suggest that cancer patients may incur higher palliative-care costs.
2. Average length of stay and share of patients with short length of stay (less than 7 days): Multiple papers in the palliative-care literature suggest hospice visits and resource-use follow a U-shaped pattern with respect to length of stay, i.e. patients receive more visits at the beginning and end of their stays than in the middle. This may mean that resource-use *during* a visit may be higher for patients at the beginning and end of their stays; if that is the case, then hospices with mostly short-stay patients may have higher per-visit costs.
3. Share of non-routine care days: These days may be more resource-intensive, and thus hospices with a higher share of non-routine care days may have higher per-visit costs.
4. Share of home-based patients: There are conflicting suggestions on whether patients in their own homes incur higher or lower costs than patients located elsewhere (such as nursing homes). Nursing-home-based patients may have lower costs since they already receive some care from the nursing home staff, but it may also mean they do not have a primary caregiver at all times, and so may lack care.

¹⁵If exit-selection effect were present, then even after conditioning on entry before 2002, there would be a “survivors’ bias” (Backus (2020)) – firms with high cost shocks and facing greater competition will exit. Thus, even for hospices with entry time before 2002, we will be left with a selected sample where higher competition markets have hospices with lower costs. Without exit-selection effect, there is no such bias.

Even after including these covariates in Equation 3, the results are qualitatively the same as before; see Tables 13 and 19.

I check if these patterns can be explained by studying the skill-mix of visits across for-profits and later entrants. I define the “skill mix of visits” as the share of visits made by higher-skilled workers (RNs, LVNs, and physicians) out of all visits made by RNs, LVNs, physicians, and home health aides, i.e. it tries to capture if a hospice may be using home health aides to substitute for clinical staff visits.

I rerun Equation 3 with skill mix of visits as an additional control, and find that the results are qualitatively the same. Hospices with higher skill mix do have higher per-visit costs, and including the variable slightly reduces the magnitude of the for-profit’s cost advantage.

I perform a variety of robustness checks. Robustness checks using alternative measures of entry time are reported in Section E. My results are also robust to not using an instrumental variables approach, see Table 17. The results are also robust to controlling for year fixed effects and therefore not driven by secular trends in any variables; see Table 48.

I also perform robustness checks by measuring competition level with number-equivalent-HHI and find similar results; see Section K.1.

While I leave out LA county for my primary analysis due to allegations of fraud, I rerun my main regressions including LA county in the data and find similar results (Section K).

I also rerun my main regressions for markets with fewer than 50 firms, and markets with fewer than 30 firms, in case there are worries about fraud in other large markets. I continue to find similar results; see Section K.

6 Discussion

In this section, I discuss mechanisms and possible confounders, and present some additional evidence. First, I discuss the validity of my instrumental variable. Second, I decompose the cost components to see which components are primarily declining in the face of higher competition.

Table 13: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.065 (0.033)	0.066* (0.033)	0.079* (0.039)	0.064* (0.032)
For-profit	-0.156** (0.048)	-0.158** (0.048)	-0.159** (0.046)	-0.158** (0.049)
Entry time	-0.016*** (0.004)	-0.016*** (0.004)	-0.017*** (0.003)	-0.015*** (0.004)
Entry before 2002	-0.032 (0.042)	-0.028 (0.042)	-0.031 (0.043)	-0.040 (0.044)
Share of patients w/ cancer	0.253* (0.118)	0.258* (0.119)	0.244* (0.120)	0.262* (0.114)
Share of patients w/ <7 days LOS	-0.138 (0.116)	-0.134 (0.114)	-0.100 (0.111)	-0.154 (0.115)
Average length of stay	-0.001 (0.001)	-0.001 (0.001)	-0.001* (0.001)	-0.001 (0.001)
Share of non-routine care days	0.041 (0.299)	0.116 (0.288)	0.397 (0.291)	0.102 (0.290)
Share of home-based patients	0.066 (0.037)	0.065 (0.038)	0.045 (0.042)	0.071 (0.036)
Total patient days / 1000		-0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.044*** (0.010)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.026* (0.010)	-0.019 (0.012)	0.015 (0.013)	-0.026* (0.010)
Inpatient unit services				-0.067 (0.038)
Pediatric program services				-0.037 (0.048)
Bereavement services				0.060** (0.022)
Adult day care services				0.268** (0.097)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of days of market-level-mean per-visit costs.

Third, reduction in per-visit costs does not appear to cause a decline in quality of care. Finally, I discuss the economic mechanisms that may be driving my results.

6.1 Discussion on instrumental variable

Empirically, my measure of expected deaths is highly correlated with population levels in the market, which leads to several potential criticisms of my instrumental variable strategy.

First, it may be the case that greater population leads to greater population density, which may mean patients live closer together, which may mechanically reduce per-visit costs as hospice staff do not have to travel very far. Lower travel time could mean less cost on transportation, or that the same nurse can make more visits each day. Thus, the exclusion restriction of the instrumental variable is violated, as expected deaths may influence per-visit costs independently of affecting competition. Second, increasing population may lead to increases in labor supply and therefore lower wages, which may also reduce per-visit costs. This would mean population/market size would reduce per-visit costs independently of affecting competition, similarly violating the exclusion restriction.

While the county fixed effects absorb time-invariant differences in population across counties, it does not account for changes in population density within counties over time. I also cannot separately control for population density in the regression because it is highly collinear with my instrumental variable.

Instead, I perform several checks to rule out this mechanism. First, recall that I found incumbents do not experience declines in per-visit costs as competition rises. This is not what we would expect if the above channels were present – rising population should reduce per-visit costs for all firms in the market, including incumbents.

Second, I run several descriptive regressions. I regress per-visit costs on population density, firm fixed effects and controls. This checks if, within a firm, increases in the market's population density are associated with decreasing per-visit costs, which would be evidence of the population density mechanism. Instead, I find no relationship between population density and per-visit costs.

I repeat this regression by including market size instead of population density and get very similar results.¹⁶ I find no significant effect of market size on per-visit cost, showing that the increase in population has no effect either through increasing competition or via any additional channels. See Appendix L.1 for results.

Also note that I control for trends of per-visit costs through including polynomials of lags of market-level-mean per-visit costs.

6.2 Decomposing cost components

Since my dataset contains breakdown of firm costs, I examine which cost components are primarily driving the results. That is, as level of competition increases, which cost components are most responsible for the decline in per-visit costs?

In Section H, I run the regression in Equation 3 on each cost component (divided by number of visits). I find that for-profits have lower costs in “nursing care”, “medical social services expenses”, and “other visiting services”. Later entrants have lower costs in “nursing care”; and higher costs in “medical supplies” and “physician services”.¹⁷

6.3 Suggestive Evidence on Quality and Reallocation

While the focus of this paper is on hospice production costs, a natural question to ask is whether hospices with lower per-visit costs also provide lower quality of care. I perform a series of empirical checks that offer no suggestive evidence of a decline in quality.

¹⁶In this case, the entry time and for-profit status are absorbed in the firm fixed effect. The coefficient on market size subsumes two mechanisms – the indirect effect on per-visit cost through shifting firm count (i.e. incumbent-treatment effect and exit-selection effect), and the direct effect of market size on per-visit cost (i.e. the additional channels above). Each of these effects should be negative. Incumbent-treatment and exit-selection effects are negative as described previously. The effect of higher population density and higher labor supply should lower per-visit costs.

¹⁷There are two points to note regarding interpreting the above results. First, I mention only outcomes which are statistically significant at 10% level or less. This should not be judged as inference on individual cost components, but rather as suggestive evidence on which cost components show a “strong” correlation. The regressions are meant to be descriptive, not causal, and do not make use of Multiple Hypothesis Testing corrections as a result. Second, firms may be trading off one cost component with another; for example, a firm may be reducing “nursing care” while raising “medical supplies” costs; this also makes interpretation of the individual cost-component regressions difficult.

First, I show that hospices with lower per-visit costs provide more visits-per-patient-day. I regress visits-per-patient day on per-visit costs and a variety of controls (Table 20), and find a strong negative effect. Visits-per-patient-day is a good proxy for the quality and effort exerted by the hospice. A hospice that visits its patients more often are checking up on them regularly and adjusting for day-to-day difficulties. I find strong evidence throughout my data that hospices that make more visits-per-patient-day also have greater market share (Table 31). I have previously shown that hospices in more competitive markets also make more visits-per-patient-day. Furthermore, in more competitive areas, market share reallocates towards higher visits-per-patient-day hospices (Figure 17 and Table 22). This suggests that patients value visits-per-patient-day.

These results suggest that hospices with lower per-visit costs are taking advantage of their lower costs to provide more visits-per-patient-day. This makes sense because there is no price competition between hospices, so consumers should choose hospices based on quality. Thus, with more firms, the quality competition should intensify, and only firms with lower per-visit costs can afford to make more visits.¹⁸

Second, I check if higher per-visit cost hospices also attract more patients – this would suggest that higher cost hospices are more attractive to patients possibly due to higher quality. To do so, I run the following descriptive regression on patient volume. I regress patient volume on visits-per-patient-day and per-visit costs (while controlling for hospice age, for-profit status, and other factors). I find that per-visit costs have no statistically significant effect (Table 31). This also suggests that hospices with higher per-visit costs are not necessarily attracting more patients, suggesting that per-visit costs are not proxying for quality.

That said, this is a descriptive regression, and it is possible that for-profit status and age are absorbing meaningful variation in the per-visit cost variable. In particular, the for-profit status

¹⁸An alternative argument could be that hospices which spend little within a visit worsen the health of their patient, and so has to make up with lots more visits. I believe this is unlikely. Visits are the most costly part of hospice care, as it involves hiring nurses, paying for their travel, and the opportunity cost of their time to the hospice. Thus, it is unlikely that a hospice is incurring the cost of a visit and then shirking on care quality during that visit. It is also not economical to shirk on within-visit care quality and then make up with more visits, since the type of care provided during visits is low-skill, while the overall cost of a visit is very high. While it may be true of a few hospices, it is unlikely to be a widespread phenomenon.

variable has a strong negative effect on patient volume. However, note that the following is still true – after controlling for for-profit status and age, higher per-visit costs do not lead to higher patient volume. Thus, after controlling for for-profit status, changes in per-visit costs do not appear to affect patient volume, which may suggest they don’t reflect quality differences.

Third, the institutional details suggest that a widespread quality reduction is unlikely. Since hospice care is effectively free for the vast majority of patients, hospices compete primarily on quality and reputation. This quality competition is more fierce in more competitive markets. Since low cost hospices generally enter more competitive markets, it would suggest they would have an incentive to raise quality, not lower it.

On reflection, it is difficult to measure if hospices are making small sacrifices in quality, and if so by how much. Part of the story for cost reduction is likely coming from hospices improving coordination, adopting better technology, improving management practices, reducing waste, and other efficiency improvements that do not affect quality. However, it is possible that some cost-cutting measures may be affecting quality in ways that are hard for me to detect.

It is also possible that the any decline in quality from cost reductions may matter little to patients. Hospices compete primarily over quality and reputation, and in more competitive markets, that competition is more intense. Thus, it is unlikely that the lower cost hospices are all sacrificing quality significantly, because they would lose market share to higher-quality rivals. Instead, it may be that patients do not value certain dimensions of within-visit quality as much, and so hospices may be cutting costs in those dimensions. The hospice may use those cost savings to provide more visits-per-patient-day, which patients do value highly according to data. This may explain why per-visit costs do not correlate positively with patient volume in Table 31.¹⁹

For instance, when I regress “skill mix of visits” on a variety of covariates (Section G), I find that later entrants are not more likely to choose a higher skill mix of visits, and for-profits are more

¹⁹This is similar to the ideas in [Cochrane \(2015\)](#): “My example industries do not cut costs by selling shoddy products or service. Instead, they provide consistent quality on the dimensions people turn out to really care about, and save on those that people don’t really care about...How can health care emulate the quality improvement and cost reductions of these successful service oriented industries? My examples share a common thread: Intense competition. And, in particular, competition from new entrants, who put old companies out of business or force unwelcome and disruptive changes.”

likely to choose a lower skill mix. Does this necessarily mean a decline in quality? No – it may be that for-profits are using lower-skill workers (such as home health aides) to perform tasks that do not require clinical skill, while reserving clinical workers for medical tasks. Thus, the hospice may be optimizing its skill mix of visits to reduce costs without sacrificing quality.

To emphasize, this project is concerned about production costs in healthcare provider markets and how they evolve with entry-exit dynamics. If production costs are lowered by firms sacrificing quality, it does not go against the main findings of this paper. That said, this section provides suggestive evidence that quality is not being sacrificed to lower production costs.

6.4 Reallocation

As competition increases, is there reallocation from high cost hospices to low cost hospices? To study this, I regress share-weighted per-visit costs on number of firms. The share-weighted per-visit cost in a market-year is calculated by weighting each hospice’s per-visit cost by its within-hospice-market-share in that county-year. Thus, if there is reallocation towards low-cost firms as competition increases, then we should see a negative relationship between share-weighted per-visit costs and number of firms.

I find a strong negative effect, i.e., market share gets reallocated towards low-cost firms as competition increases (Table 14). I plot the relationship in Figure 10 to aid visualization.

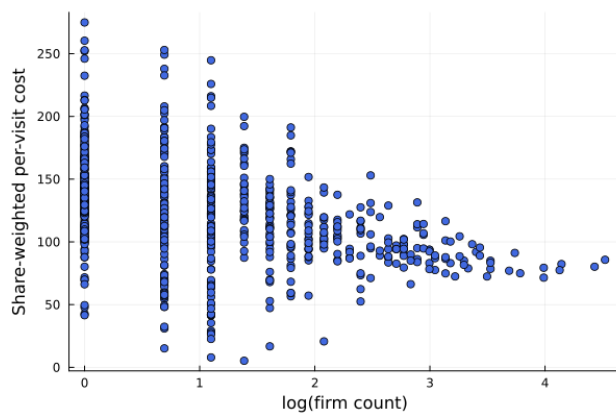
What is the process by which market share gets allocated to lower-cost hospices? I argue that lower-cost hospices make more visits-per-patient-day (as shown earlier), and thus provide higher quality of care. Patients then choose these hospices, leading to higher market share. Table 31 regresses patient volume on visits-per-patient-day and per-visit costs (while controlling for hospice age, for-profit status, and other factors), and finds that visits-per-patient-day strongly correlate with volume, but per-visit costs have no statistically significant effect. This suggests that patients are choosing hospices based on visits-per-patient-day rather than per-visit costs (but see the caveats in the previous discussion).

Table 14: Regression of share-weighted per-visit costs vs. number of firms

	Share-weighted per-visit cost	Share-weighted log per-visit cost
	(1)	(2)
log(firm count)	-17.261*** (1.124)	-0.331*** (0.021)
Year Fixed Effects	Yes	Yes
Obs.	700	700

Notes: Robust standard errors in parentheses.

Figure 10: Scatter of share-weighted per-visit cost in a market-year by number of hospices.



Notes: Each point represents a county-year. The share-weighted per-visit cost is calculated for a county-year by weighting each hospice's per-visit cost by its within-hospice-market-share in that county-year.

6.5 Discussion on economic mechanisms behind firm decisions

In this section, I discuss possible economic mechanisms that may be driving my results, and provide some additional empirical evidence.

First, there is the question of how some hospices can attain cost reductions, apart from potential quality reductions (discussed elsewhere). Reading about hospice management online suggests it is a mix of improving coordination, monitoring and tracking costs, limiting unnecessary resource use or waste, adopting improved technology, intense bargaining over input prices, better management practices, etc.

What explains why later entrants have lower per-visit costs than past entrants? My preferred explanation is that with greater competition, firms need to compete more on quality (i.e. visits-per-patient-day), and increased visits-per-patient-day is more costly to provide. Only firms with lower per-visit costs can afford to make more visits while still being profitable. Thus, as competition goes up, the threshold level of cost that firms need to be below to enter the market goes down. This explains why earlier firms have higher costs, and per-visit costs decline with entry time. An alternative explanation on a similar note is that when considering entry into a market with already several hospices, an entrant will anticipate strong competition over patients, and therefore will need to have lower costs to maintain enough profits to justify incurring the entry cost.

It is also surprising that incumbents do not reduce costs when facing greater competition. I speculate that this has to do with organizational or managerial inertia that prevents incumbents from adjusting their cost structures.

Another puzzling fact is the absence of exit-selection effect. An explanation could be that the market for hospices is growing (rapidly in some counties), and such market growth may be sufficient to accommodate new and existing firms, even those with high costs.

Why do non-profits have higher costs than for-profits? There could be multiple reasons. The first is intrinsic motivation and salience – for-profits want to maximize profits, and so focus more on cost cutting; non-profits may have more altruistic objectives and have less pressure on maximizing profits. Second, non-profits may be partly funded by donations and enjoy tax advantages, and so

they are less concerned about lowering costs to justify their operation.

To examine how important donations may be for non-profits, I study donations-per-patient of non-profits in my data. Table 15 plots the distribution of donations-per-patient at the hospice-year level. Surprisingly, I find that 40% of non-profit hospice-years have zero donations-per-patient. For more than 30% of non-profit hospice-years, donations-per-patient is greater than \$300. This is a sizeable amount, as the median per-visit cost is around \$100.

I also find that non-profits appear to have only a small tax advantage over for-profits. I study the distribution of taxes-per-patient for for-profits and non-profits. As expected, non-profits do not pay taxes in over 96% of observations. Surprisingly, over 70% of for-profit hospice-years also do not report paying taxes. Of the remainder that do, there is large variation ranging from \$18 per patient to hundreds of dollars per patient.

In Section B, I rule out the possibility that my finding of cost differences between non-profits and for-profits is driven by misreporting from non-profits.

7 Conclusion

I study the effect of entry-exit dynamics on hospice visit costs, and how it is influenced by the presence of non-profit and for-profit firms. Answering this will shed light on how competition may improve economic efficiency and lower rising healthcare costs in the United States. I use a panel dataset of hospices in California for 2002-2018 to construct hospice-level measures of per-visit costs and competition in markets. Using linear and quantile regressions with instrumental variables, I show that per-visit costs are lower in counties with more hospices.

What drives the negative relationship between per-visit costs and competition? First, for-profits have lower costs than non-profits, and the entry of for-profits helps drive down overall costs in the market. Second, I show that entrants have lower costs than incumbents, and successive entrants have lower costs than prior entrants. Third, I show that higher-cost incumbents are more likely to exit the market. Fourth, I show that greater competition does not lead to incumbents lowering their

per-visit costs or for higher-cost incumbents to exit the market, and that this is robust to controlling for patient characteristics. I find no evidence that reduction in per-visit costs leads to a decline in the quality of care.

The key policy insight is that entry barriers – such as the recently implemented Certificate-of-Need laws for hospices in California – can impede cost reductions from competition and also worsen patient access to care, since for-profits have improved access for patients.

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Online Appendix

A Estimation details

A.1 Quantile regression

This section follows similar descriptions in [Larsen et al. \(2023\)](#).

The grouped-quantile IV regression estimator closely follows [Backus \(2020\)](#), and has also been used in [Chetverikov et al. \(2016\)](#) and [Larsen et al. \(2023\)](#). It is appropriate for settings with group-level treatment (in my case, county-year competition) and micro data on outcomes within a group (in my case, firm-level per-visit costs).

In my setting, a group is a county-by-year cell, the treatment variable is county-by-year competition level, and micro data on outcomes corresponds to individual hospices in each county-year.

Since my quantile analysis focuses on deciles (10th percentile up to 90th percentile), this raises the question of how I calculate quantiles/deciles for markets with fewer than 10 firms. I use 2 methods to get around this:

1. I rerun the grouped-quantile IV regression but restrict the sample to county-years with at least 10 firms. The results are qualitatively the same.
2. I use the programming language *Julia*'s default setting for calculating quantiles, which uses linear interpolation between points following [Hyndman and Fan \(1996\)](#). This allows it to calculate quantiles even when there are fewer than 10 firms in a county-year.²⁰ See Figure

²⁰The following examples serve as illustration of how *Julia* interpolates. For the vector [1,2,3,4,5], *Julia* calculates the 0th quantile to be 1, the 10th quantile to be 1.4, and the 25th quantile to be 2. For the vector [1, 2, 3], *Julia* calculates the 0th quantile to be 1, the 10th quantile to be 1.2, the 25th quantile to be 1.5, and the 50th quantile to be 2.

11 to see the exact description from Julia’s documentation.

Figure 11: Julia documentation for calculating the p th quantile via interpolation.

Samples quantile are defined by $Q(p) = (1-\gamma)*x[j] + \gamma*x[j+1]$, where $x[j]$ is the j -th order statistic of v , $j = \text{floor}(n*p + m)$, $m = \alpha + p*(1 - \alpha - \beta)$ and $\gamma = n*p + m - j$.

By default ($\alpha = \beta = 1$), quantiles are computed via linear interpolation between the points $((k-1)/(n-1), x[k])$, for $k = 1:n$ where $n = \text{length}(v)$. This corresponds to Definition 7 of Hyndman and Fan (1996), and is the same as the R and NumPy default.

The keyword arguments α and β correspond to the same parameters in Hyndman and Fan, setting them to different values allows to calculate quantiles with any of the methods 4-9 defined in this paper:

- Def. 4: $\alpha=0, \beta=1$
- Def. 5: $\alpha=0.5, \beta=0.5$ (MATLAB default)
- Def. 6: $\alpha=0, \beta=0$ (Excel PERCENTILE.EXC, Python default, Stata `altdf`)
- Def. 7: $\alpha=1, \beta=1$ (Julia, R and NumPy default, Excel PERCENTILE and PERCENTILE.INC, Python 'inclusive')
- Def. 8: $\alpha=1/3, \beta=1/3$
- Def. 9: $\alpha=3/8, \beta=3/8$

Notes: Source url - <https://docs.julialang.org/en/v1/stdlib/Statistics>

I regress market-level quantiles on competition level in the market. I instrument for competition level with market size, i.e. expected number of deaths. In additional robustness checks, I include second-order polynomials of one- and two-year lags of county-level mean of per-visit costs. I include county fixed effects in all regressions (similar to [Larsen et al. \(2023\)](#), who incorporate state fixed effects in their regressions). Note that every parameter is estimated separately for each quantile.

[Backus \(2020\)](#) finds that such regressions suffer from an “order-statistic bias”, which derives from the fact that some of the groups (county-years) have few firms present. He shows that the order-statistic bias is a nonlinear function of the number of firms, and thus including a flexible function of the number of firms in the grouped quantile regression fixes the bias. Following his method, I use third-degree polynomials of the number of firms to overcome the order-statistic bias.

With all this in place, I estimate the following grouped-quantile IV regression. Let k denote the k th decile, and $\rho_{mt}^{(k)}$ be the k th decile of the market-level distribution of per-visit costs.

$$\rho_{mt}^{(k)} = \alpha_m^{(k)} + \alpha_f^{(k)} f_{mt} + X_{mt}' \alpha_x^{(k)} + g^{(k)}(n_{mt}) + v_{mt}$$

where f_{mt} is the level of competition and n_{mt} is the number of firms in the market. The semipara-

metric correction for the order statistic bias, $g^{(k)}(\cdot)$, is a third-degree polynomial of the number of firms in the market, and is estimated separately for each k .

Similar to [Backus \(2020\)](#) and [Larsen et al. \(2023\)](#), estimating this involves a simple linear-regression (with IV) on market-level data for each quantile.

A.2 Smoothed density plots

Throughout the paper, I use smoothed density plots to visualize distributions of variables. These are kernel density plots, and for bandwidth use Silverman's rule.

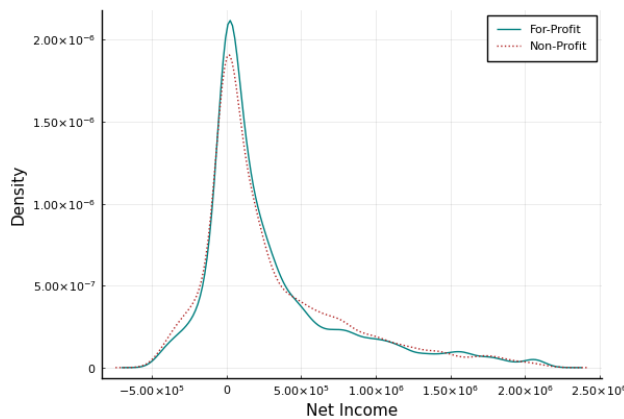
B Misreporting of costs by non-profits

One potential worry is that my finding of cost differences between non-profits and for-profits may be driven by misreporting from non-profits. If non-profits overstate their costs, then the true cost differences between non-profits and for-profits may be smaller than what I find. This may happen if non-profits have incentives to appear less profitable in order to receive donations or avoid scrutiny. (Interestingly, it has also been suggested that non-profits may have incentives to understate costs to appear efficient to donors and receive more donations.)

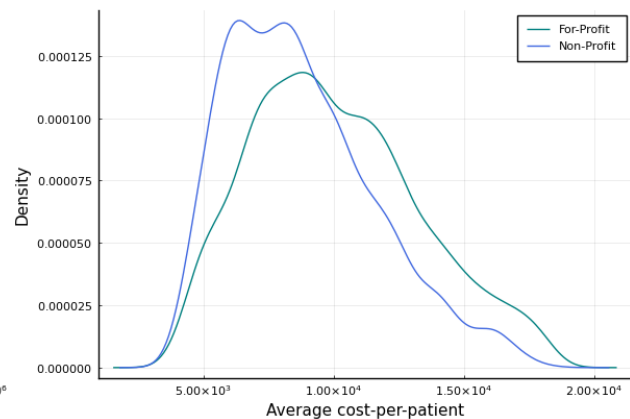
To rule out this possibility, I plot i) average cost-per-patient and ii) reported profits, of non-profits versus for-profits in Figure 12; also see Figure 13 for histograms.

I find distributions of for-profits and non-profits to be quite similar, and in particular do not find non-profits to systematically have lower profits or higher average cost-per-patient.

Figure 12: Distributions of reported profits (“net income”) and cost-per-patient by for-profit status.



(a) Smoothed kernel density of reported profits, broken down by for-profit status.



(b) Notes: Smoothed kernel density of cost-per-patient, broken down by for-profit status.

Figure 13: Histograms of reported profits (“net income”) and cost-per-patient by for-profit status.

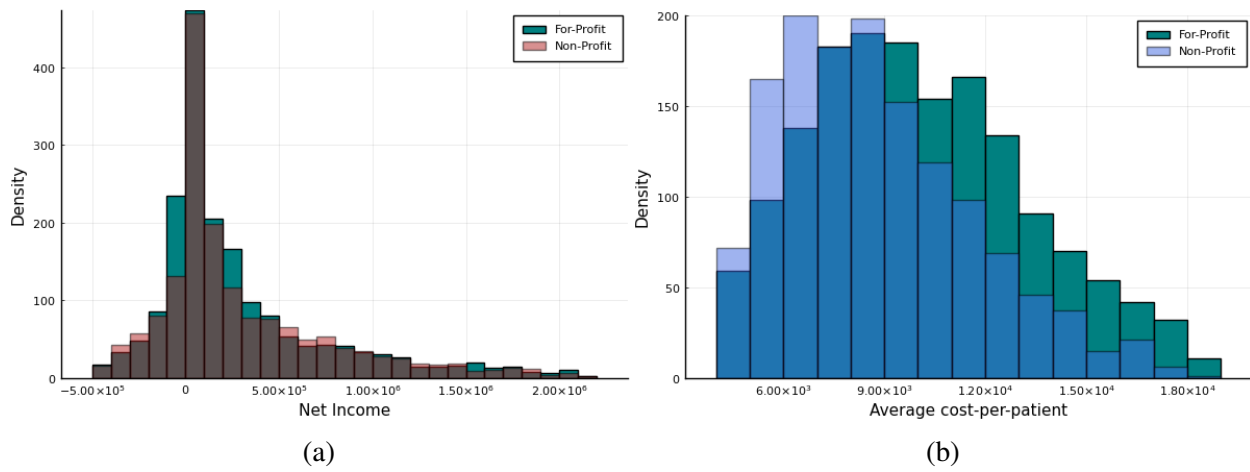


Table 15: Quantiles of donations-per-patient for non-profits.

Quantile	Donations per Patient (\$)
0.0	-41.37
0.1	0.0
0.2	0.0
0.3	0.0
0.4	13.31
0.5	93.11
0.6	227.2
0.7	382.56
0.8	676.45
0.9	1249.45
1.0	13068.4

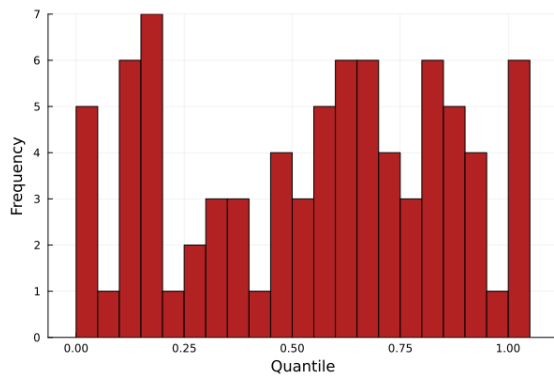
C Links to data

1. Hospice data: <https://hcai.ca.gov/data/healthcare-utilization/home-health-agency-and-hospice-utilization/>
2. 2000-2010 population data by age for California: <https://dof.ca.gov/forecasting/demographics/estimates/annual-intercensal-population-estimates-by-race-ethnicity-with-age-and-gender-detail/>
3. 2010-2020 population data by age for California: <https://dof.ca.gov/forecasting/demographics/estimates/annual-intercensal-population-estimates-by-race-ethnicity-with-age-and-gender-detail-2010-2020/>
4. Death rates by sex for years 2002-2018 for USA: <https://www.ssa.gov/oact/HistEst/Death/2023/DeathProbabilities2023.html>

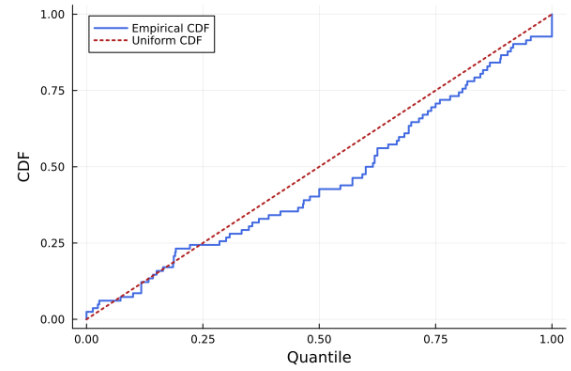
The population data comes from the State of California Department of Finance and the US Census. The death rates come from the Social Security Administration.

D Miscellaneous

Figure 14: Distribution of per-visit costs of exiting hospices



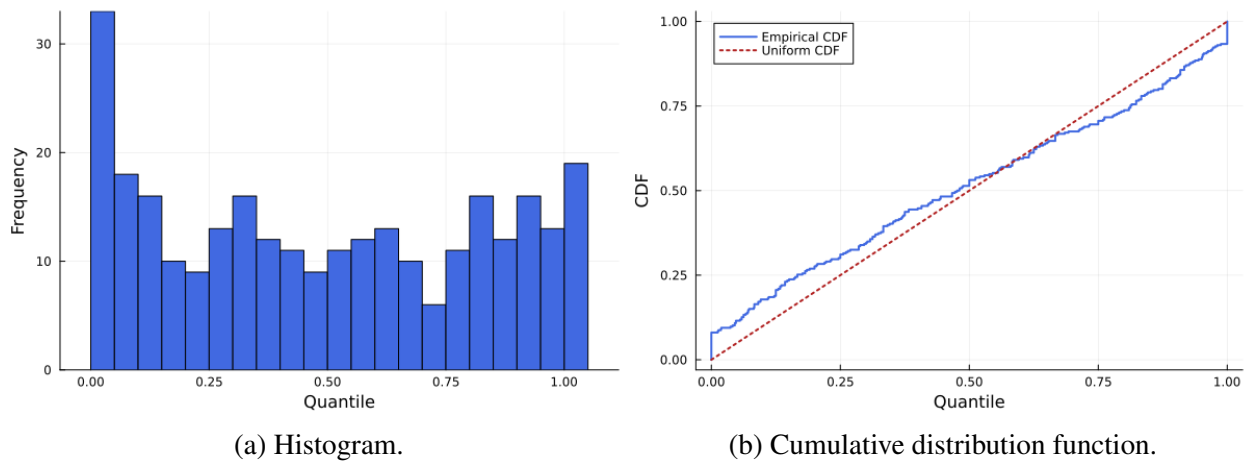
(a)



(b)

Notes: Figures 9a and 9b plot the quantiles of per-visit costs of exiting hospices in a county-year, using leave-one-out. These plots limit to markets with greater than 10 hospices for reliable quantile calculation.

Figure 15: Quantiles of per-visit costs of entrants in a county-year, using leave-one-out.



Notes: These plots limit to markets with greater than 10 hospices for reliable quantile calculation.

Table 16: First stage of IV regression

	log(firm count)			
	(1)	(2)	(3)	(4)
log(market size)	4.611*** (0.651)	4.606*** (0.651)	4.607*** (0.652)	4.605*** (0.645)
Share of patients w/ cancer	-0.064 (0.066)	-0.065 (0.067)	-0.063 (0.065)	-0.059 (0.071)
Share of patients w/ <7 days LOS	-0.106 (0.111)	-0.108 (0.112)	-0.108 (0.112)	-0.104 (0.111)
Average length of stay	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Share of non-routine care days	-0.142 (0.205)	-0.174 (0.208)	-0.167 (0.192)	-0.140 (0.194)
Share of home-based patients	-0.019 (0.034)	-0.018 (0.034)	-0.017 (0.033)	-0.022 (0.033)
For-profit	0.054 (0.028)	0.055 (0.028)	0.054 (0.028)	0.051 (0.026)
Entry time	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)
Entry before 2002	0.126* (0.051)	0.126* (0.051)	0.126* (0.051)	0.131* (0.052)
Total patient days / 1000		0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			0.003 (0.007)	
Total Visits / 10,000 squared			-0.000 (0.000)	
log(operating cost)	0.004 (0.007)	0.000 (0.008)	0.001 (0.011)	0.005 (0.007)
Inpatient unit services				-0.015 (0.031)
Pediatric program services				0.005 (0.026)
Bereavement services				-0.018 (0.011)
Adult day care services				-0.128 (0.175)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 17: Cost-competition regression without instruments (OLS regression).

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.030 (0.021)	0.031 (0.021)	0.038 (0.020)	0.031 (0.020)
For-profit	-0.153** (0.048)	-0.154** (0.048)	-0.156** (0.046)	-0.155** (0.049)
Entry time	-0.014*** (0.004)	-0.014*** (0.003)	-0.015*** (0.003)	-0.013*** (0.003)
Entry before 2002	-0.025 (0.043)	-0.021 (0.043)	-0.023 (0.044)	-0.033 (0.045)
Share of patients w/ cancer	0.244* (0.119)	0.249* (0.121)	0.234 (0.122)	0.255* (0.115)
Share of patients w/ <7 days LOS	-0.159 (0.117)	-0.156 (0.115)	-0.125 (0.113)	-0.174 (0.115)
Average length of stay	-0.001 (0.001)	-0.001 (0.001)	-0.001* (0.001)	-0.001 (0.001)
Share of non-routine care days	0.029 (0.301)	0.101 (0.290)	0.380 (0.289)	0.092 (0.292)
Share of home-based patients	0.066 (0.037)	0.064 (0.038)	0.045 (0.042)	0.071 (0.036)
Total patient days / 1000		-0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.044*** (0.010)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.025* (0.010)	-0.018 (0.011)	0.016 (0.012)	-0.025* (0.009)
Inpatient unit services				-0.068 (0.039)
Pediatric program services				-0.037 (0.048)
Bereavement services				0.059* (0.022)
Adult day care services				0.261* (0.099)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at the county level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 18: Regression of per-visit costs on competition and for-profit status, limited to hospices that entered on or before 2002.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.079 (0.066)	0.072 (0.066)	0.064 (0.064)	0.087 (0.066)
For-profit	-0.099 (0.055)	-0.098 (0.054)	-0.107 (0.054)	-0.086 (0.051)
Total patient days / 1000		-0.003 (0.001)		
Total patient days / 1000 squared		0.000** (0.000)		
Total Visits / 10,000			-0.023 (0.013)	
Total Visits / 10,000 squared			0.000 (0.000)	
log(operating cost)	-0.068*** (0.015)	-0.048 (0.024)	-0.030 (0.025)	-0.084*** (0.017)
Inpatient unit services				0.026 (0.045)
Pediatric program services				0.135* (0.065)
Bereavement services				0.059 (0.043)
Adult day care services				0.365*** (0.058)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	891	891	891	891

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 19: Regression of per-visit costs on competition and for-profit status, limited to hospices that entered on or before 2002. Includes patient characteristics as controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.100 (0.079)	0.097 (0.079)	0.092 (0.077)	0.104 (0.081)
For-profit	-0.060 (0.051)	-0.061 (0.052)	-0.073 (0.051)	-0.050 (0.049)
Share of patients w/ cancer	0.413 (0.241)	0.407 (0.253)	0.445 (0.264)	0.365 (0.239)
Share of patients w/ <7 days LOS	-0.429* (0.195)	-0.402* (0.198)	-0.405 (0.210)	-0.430* (0.189)
Average length of stay	-0.004** (0.001)	-0.004* (0.001)	-0.003* (0.001)	-0.004** (0.001)
Share of non-routine care days	0.247 (0.740)	0.411 (0.642)	0.737 (0.680)	0.232 (0.851)
Share of home-based patients	0.088 (0.110)	0.084 (0.112)	0.065 (0.108)	0.096 (0.112)
Total patient days / 1000		-0.002 (0.001)		
Total patient days / 1000 squared		0.000 (0.000)		
Total Visits / 10,000			-0.022 (0.014)	
Total Visits / 10,000 squared			0.000 (0.000)	
log(operating cost)	-0.062** (0.018)	-0.049 (0.026)	-0.028 (0.026)	-0.075*** (0.020)
Inpatient unit services				0.004 (0.043)
Pediatric program services				0.109 (0.057)
Bereavement services				0.068 (0.043)
Adult day care services				0.375*** (0.063)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	891	891	891	891

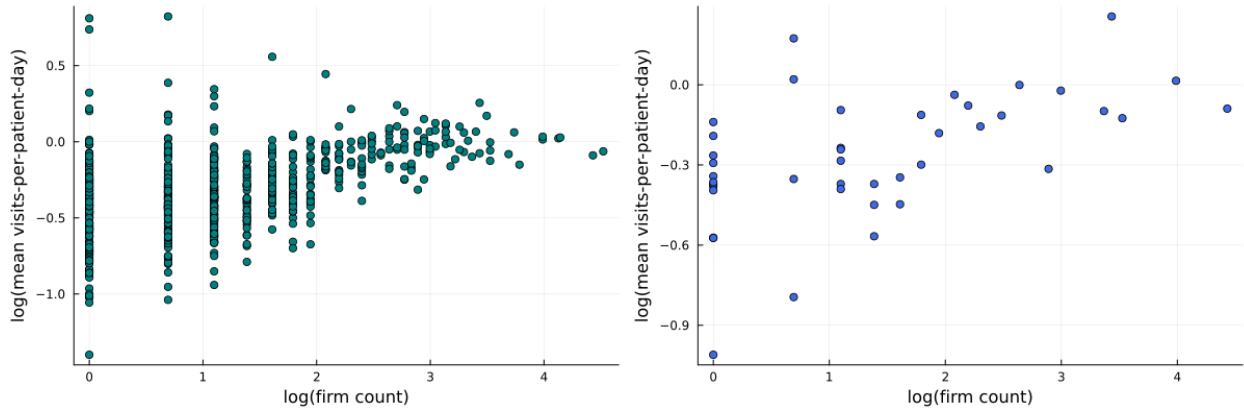
Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 20: Regression of visits-per-patient-day on per-visit costs and for-profit status.

	Visits-per-patient-day
log(per-visit cost)	-0.214*** (0.015)
For-profit	-0.005 (0.022)
County Fixed Effects	Yes
Controls	Yes
Obs.	3,155

Notes: This is limited to observations with visits-per-patient-day below 1.5 to remove outliers. Standard errors clustered at county-level in parentheses. Controls are log of firm count, hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, daycare for adult services indicator, for-profit status, and agency types indicators.

Figure 16: Variation in mean visits-per-patient-day over time and space.



(a) Plot of county-level-mean of visits-per-patient-day over the full dataset.

(b) Plot of county-level-mean of visits-per-patient-day for 2017.

Table 21: Regression of county-level-mean visits-per-patient-day on number of hospices.

	Mean visits-per-patient-day	log(mean visits-per-patient-day)
	(1)	(2)
log(firm count)	0.086** (0.028)	0.120** (0.036)
County Fixed Effects	Yes	Yes
Obs.	699	699

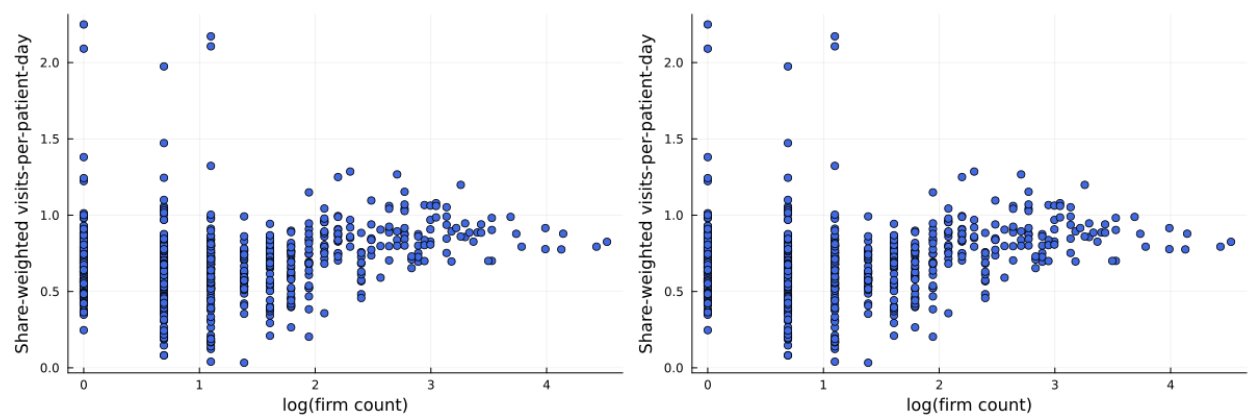
Notes: Each observation is a county-year. For the county-year, I calculate the mean visits-per-patient-day by averaging over all hospices in that county-year. Standard errors clustered at the county level in parentheses.

Table 22: Regression of share-weighted visits-per-patient-day on number of hospices.

	Share-weighted visits-per-patient-day	Share-weighted log visits-per-patient-day
	(1)	(2)
log(firm count)	0.047*** (0.008)	0.118*** (0.008)
Year Fixed Effects	Yes	Yes
Obs.	700	700

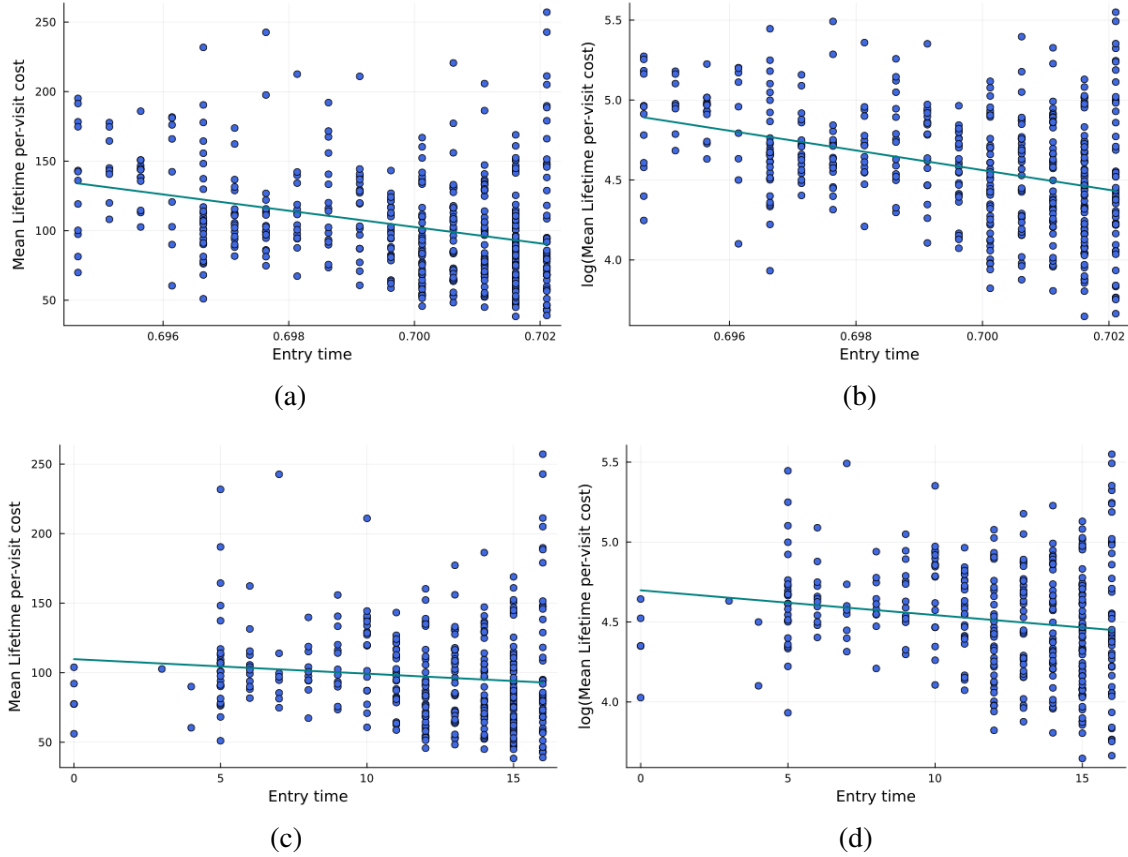
Notes: Each observation is a county-year. Standard errors clustered at the county level in parentheses.

Figure 17: Scatter of share-weighted visits-per-patient-day against firm count.



County	Total Exits	Total Entries	Total Unique Hospices
San Bernardino	32	117	124
Orange	25	76	88
Ventura	12	47	52
Riverside	13	38	47
San Diego	6	26	37
Sacramento	7	25	30
Alameda	6	19	26
Contra Costa	10	18	22
Santa Clara	6	16	22
San Mateo	4	13	16
Kern	5	11	14
Fresno	3	10	13
San Joaquin	4	10	12
Santa Barbara	3	8	10
Sonoma	5	7	11
Placer	2	5	7
San Francisco	5	5	11
Shasta	2	5	6
Marin	3	4	5
Stanislaus	2	4	6
Sutter	3	4	6
Mariposa	3	3	4
Santa Cruz	1	3	4
Tulare	1	3	4
Merced	2	2	4
Tehama	1	2	2
El Dorado	1	1	3
Kings	1	1	2
Mendocino	1	1	2
Monterey	1	1	4
Napa	1	1	2
Siskiyou	2	1	3
Yolo	1	1	2
Butte	1	0	4
Del Norte	1	0	1
Madera	1	0	1

Figure 18: Scatterplot of mean per-visit costs versus entry time.



Notes: Each point is a unique hospice. The mean is calculated over the lifetime of the hospice, i.e. the mean per-visit cost for a specific hospice over all years observed in the dataset. Entry time is measured as number of years since 2002.

Figure 19: Snapshot of the “operating expense form”.

DETAIL OF OPERATING EXPENSES

(do not input “\$” signs, commas, or decimals; round up to whole dollar)

Use data from Medicare Cost Report where applicable.

Line No.		Total (1)
	General Service Cost Centers	
30	Administrative and General	
	Inpatient Care Service	
31	Inpatient – General Care	
32	Inpatient – Respite Care	
	Program Supervision	
35	Hospice Program / Team Supervision (Non-Visit Wages)	
	Visiting Services	
36	Physician Services	
37	Nursing Care	
38	Rehabilitation Services (PT, OT, Speech)	
39	Medical Social Services – Direct	
40	Spiritual Counseling	
41	Dietary Counseling	
42	Counseling – Other	
43	Home Health Aides and Homemakers	
44	Other Visiting Services	
	Hospice Service Cost Centers	
45	Drugs, Biologicals, and Infusion	
46	Durable Medical Equipment / Oxygen	
47	Patient Transportation	
48	Imaging, Lab, and Diagnostics	
49	Medical Supplies	
50	Outpatient Services (including ER Dept.)	
51	Radiation Therapy	
52	Chemotherapy	
53	Other Hospice Service Costs	
	Other Hospice Costs	
54	Bereavement Program Costs	
55	Volunteer Program Costs	
56	Fundraising Costs	
	Other Costs	
57	Other Program Costs*	
59	Total Operating Expenses	

* Program costs including community education and outreach program costs.

Notes: This is a screenshot of the operating expense form that hospices in California have to fill out each year for HCAI. Data from this form is used to compile the dataset used in this paper.

Table 23: Percent of visits by care type.

Variable	10%	25%	50%	75%	90%
Routine care	97.59	99.12	99.69	99.91	100.0
Inpatient care	0.0	0.0	0.07	0.37	1.13
Respite care	0.0	0.0	0.06	0.18	0.38
Continuous care	0.0	0.0	0.01	0.1	0.74

Table 24: Regression of exit including interaction of cost with competition.

	Exit
log(per-visit cost)	-0.004 (0.027)
log(firm count)	-0.065 (0.033)
log(patients)	-0.030*** (0.006)
log(per-visit cost) x log(firm count)	0.010 (0.008)
County Fixed Effects	Yes
Obs.	2,896
R-squared	0.029
Adj. R-squared	0.013

E Robustness checks with alternative entry time measures

For this section, “Entry order” is defined as the order of entry for hospices after the year 2002 for a given county. For example, suppose a county had no hospices before 2002. The first hospice to enter a county after 2002 has entry order 1, the second hospice to enter has entry order 2, and so on. Hospices entering in the same year are assigned the same entry order. If a county had hospices before 2002, then all those hospices are assigned entry order 1.

Table 25: Entry time = entry order

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.083*	0.082*	0.098*	0.082*
	(0.033)	(0.033)	(0.042)	(0.032)
For-profit	-0.243***	-0.243***	-0.247***	-0.241***
	(0.041)	(0.042)	(0.039)	(0.043)
Entry time	-0.018**	-0.018**	-0.021**	-0.018**
	(0.006)	(0.006)	(0.006)	(0.005)
Entry before 2002	0.036	0.039	0.036	0.029
	(0.039)	(0.039)	(0.044)	(0.038)
Total patient days / 1000		0.000		
		(0.001)		
Total patient days / 1000 squared		-0.000		
		(0.000)		
Total Visits / 10,000			-0.048***	
			(0.013)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.022*	-0.018	0.026	-0.023*
	(0.009)	(0.015)	(0.015)	(0.009)
Inpatient unit services				-0.026
				(0.038)
Pediatric program services				-0.004
				(0.046)
Bereavement services				0.036
				(0.020)
Adult day care services				0.115
				(0.093)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 26: Entry time = entry order, with patient characteristics as controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.104** (0.035)	0.104** (0.035)	0.128** (0.045)	0.103** (0.034)
For-profit	-0.214*** (0.041)	-0.214*** (0.041)	-0.221*** (0.040)	-0.214*** (0.042)
Entry time	-0.018** (0.005)	-0.017** (0.005)	-0.021*** (0.006)	-0.017*** (0.005)
Entry before 2002	0.033 (0.040)	0.036 (0.040)	0.030 (0.044)	0.026 (0.039)
Share of patients w/ cancer	0.307* (0.124)	0.312* (0.124)	0.291* (0.122)	0.314* (0.120)
Share of patients w/ <7 days LOS	-0.265* (0.127)	-0.264* (0.128)	-0.217 (0.123)	-0.276* (0.126)
Average length of stay	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
Share of non-routine care days	0.224 (0.328)	0.278 (0.317)	0.618 (0.357)	0.263 (0.326)
Share of home-based patients	0.058 (0.055)	0.058 (0.056)	0.039 (0.061)	0.062 (0.056)
Total patient days / 1000		0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.050*** (0.012)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.015 (0.010)	-0.012 (0.015)	0.032 (0.016)	-0.016 (0.010)
Inpatient unit services				-0.040 (0.040)
Pediatric program services				-0.020 (0.046)
Bereavement services				0.040 (0.021)
Adult day care services				0.131 (0.098)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 27: Entry time = log(year of entry minus 2002 + 1)

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.068*	0.067*	0.081*	0.069*
	(0.032)	(0.032)	(0.039)	(0.032)
For-profit	-0.232***	-0.232***	-0.234***	-0.230***
	(0.041)	(0.041)	(0.038)	(0.042)
Entry time	-0.098**	-0.097**	-0.116***	-0.096**
	(0.034)	(0.032)	(0.030)	(0.032)
Entry before 2002	-0.072	-0.067	-0.093	-0.079
	(0.070)	(0.069)	(0.069)	(0.068)
Total patient days / 1000		0.000		
		(0.001)		
Total patient days / 1000 squared		-0.000		
		(0.000)		
Total Visits / 10,000			-0.047***	
			(0.013)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.019*	-0.015	0.029	-0.020*
	(0.009)	(0.015)	(0.016)	(0.009)
Inpatient unit services				-0.029
				(0.038)
Pediatric program services				-0.005
				(0.045)
Bereavement services				0.040
				(0.020)
Adult day care services				0.124
				(0.097)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 28: Entry time = log(year of entry minus 2002 + 1), with patient characteristics as controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.090*	0.090*	0.111*	0.090*
	(0.035)	(0.035)	(0.042)	(0.035)
For-profit	-0.203***	-0.203***	-0.208***	-0.204***
	(0.040)	(0.040)	(0.039)	(0.042)
Entry time	-0.093**	-0.091**	-0.113***	-0.090**
	(0.032)	(0.030)	(0.028)	(0.030)
Entry before 2002	-0.069	-0.063	-0.095	-0.075
	(0.069)	(0.068)	(0.067)	(0.068)
Share of patients w/ cancer	0.322*	0.326*	0.309*	0.328*
	(0.127)	(0.127)	(0.124)	(0.123)
Share of patients w/ <7 days LOS	-0.256*	-0.256*	-0.209	-0.269*
	(0.125)	(0.126)	(0.121)	(0.123)
Average length of stay	-0.002*	-0.002*	-0.002*	-0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
Share of non-routine care days	0.194	0.248	0.574	0.238
	(0.344)	(0.337)	(0.383)	(0.339)
Share of home-based patients	0.055	0.055	0.036	0.060
	(0.057)	(0.058)	(0.064)	(0.058)
Total patient days / 1000		0.000		
		(0.001)		
Total patient days / 1000 squared		-0.000		
		(0.000)		
Total Visits / 10,000			-0.049***	
			(0.012)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.012	-0.009	0.035*	-0.013
	(0.010)	(0.015)	(0.017)	(0.010)
Inpatient unit services				-0.044
				(0.040)
Pediatric program services				-0.022
				(0.045)
Bereavement services				0.043*
				(0.021)
Adult day care services				0.140
				(0.102)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 29: Entry time = log(entry order)

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.082* (0.032)	0.083* (0.033)	0.099* (0.040)	0.082* (0.032)
For-profit	-0.236*** (0.040)	-0.237*** (0.041)	-0.239*** (0.037)	-0.234*** (0.042)
Entry time	-0.122*** (0.033)	-0.122*** (0.034)	-0.147*** (0.035)	-0.119*** (0.031)
Entry before 2002	-0.067 (0.052)	-0.065 (0.052)	-0.089 (0.056)	-0.072 (0.050)
Total patient days / 1000		-0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.050*** (0.013)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.023* (0.009)	-0.017 (0.015)	0.026 (0.016)	-0.024* (0.009)
Inpatient unit services				-0.024 (0.038)
Pediatric program services				-0.004 (0.046)
Bereavement services				0.037 (0.020)
Adult day care services				0.125 (0.095)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 30: Entry time = log(entry order), with patient characteristics as controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.102** (0.034)	0.104** (0.034)	0.129** (0.043)	0.102** (0.033)
For-profit	-0.208*** (0.040)	-0.209*** (0.040)	-0.214*** (0.039)	-0.208*** (0.041)
Entry time	-0.116*** (0.031)	-0.116*** (0.031)	-0.144*** (0.033)	-0.112*** (0.029)
Entry before 2002	-0.064 (0.051)	-0.061 (0.051)	-0.092 (0.055)	-0.068 (0.049)
Share of patients w/ cancer	0.311* (0.126)	0.315* (0.127)	0.294* (0.124)	0.317* (0.123)
Share of patients w/ <7 days LOS	-0.266* (0.127)	-0.265* (0.127)	-0.219 (0.123)	-0.278* (0.126)
Average length of stay	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
Share of non-routine care days	0.235 (0.327)	0.301 (0.315)	0.643 (0.355)	0.274 (0.324)
Share of home-based patients	0.055 (0.055)	0.055 (0.056)	0.035 (0.061)	0.059 (0.056)
Total patient days / 1000		0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.052*** (0.013)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.016 (0.011)	-0.011 (0.015)	0.032 (0.017)	-0.017 (0.011)
Inpatient unit services				-0.039 (0.039)
Pediatric program services				-0.021 (0.046)
Bereavement services				0.041 (0.021)
Adult day care services				0.141 (0.100)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,093	3,093	3,093	3,093

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

F Regression on patient volume

Table 31: Regression of patient volume on covariates.

	log(patients)
log(firm count)	-0.492*** (0.107)
log(visits-per-patient-day)	0.342*** (0.067)
log(per-visit cost)	-0.020 (0.063)
For-profit	-0.448*** (0.078)
log(age)	0.538*** (0.035)
County Fixed Effects	Yes
Controls	Yes
Obs.	3,118

Notes: This is limited to observations with visits-per-patient-day below 1.5 to remove outliers. Standard errors clustered at county-level in parentheses. Controls are log of market size, hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, daycare for adult services indicator, for-profit status, and agency types indicators.

Hospice age is calculated as number of years since entry for those that entered after 2002, and number of years since 2002 for those that entered before/on 2002.

G Regression on skill mix of visits

	Skill mix of visits	
	(1)	(2)
Share of patients w/ cancer	0.052 (0.031)	0.046 (0.029)
Share of patients w/ <7 days LOS	-0.011 (0.036)	-0.005 (0.036)
Average length of stay	-0.000 (0.000)	-0.000 (0.000)
Share of non-routine care days	0.063 (0.050)	0.062 (0.050)
Share of home-based patients	0.065*** (0.011)	0.063*** (0.011)
Entry before 2002	0.001 (0.011)	0.002 (0.011)
Entry time	0.001 (0.001)	0.001 (0.001)
log(operating cost)	-0.009*** (0.002)	-0.009*** (0.002)
For-profit	-0.036*** (0.006)	-0.032*** (0.006)
log(firm count)	0.011 (0.009)	0.010 (0.009)
log(per-visit cost)		0.024* (0.010)
County Fixed Effects	Yes	Yes
Obs.	3,314	3,314

Notes: “Skill mix of visits” is the share of visits made by higher-skilled workers (RNs, LVNs, and physicians) out of all visits made by RNs, LVNs, physicians, and home health aides. Standard errors clustered at county-level in parentheses.

Table 33

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.063*	0.064*	0.077*	0.061*
	(0.031)	(0.031)	(0.037)	(0.030)
For-profit	-0.144**	-0.146**	-0.146**	-0.147**
	(0.047)	(0.047)	(0.045)	(0.047)
Entry time	-0.016***	-0.016***	-0.018***	-0.015***
	(0.004)	(0.004)	(0.003)	(0.004)
Entry before 2002	-0.033	-0.029	-0.032	-0.040
	(0.041)	(0.041)	(0.041)	(0.043)
Skill mix of visits	0.332*	0.346*	0.366*	0.342*
	(0.148)	(0.152)	(0.149)	(0.152)
Share of patients w/ cancer	0.236*	0.240*	0.224	0.248*
	(0.116)	(0.117)	(0.117)	(0.112)
Share of patients w/ <7 days LOS	-0.132	-0.127	-0.093	-0.149
	(0.112)	(0.110)	(0.106)	(0.111)
Average length of stay	-0.001	-0.001	-0.001*	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Share of non-routine care days	0.030	0.118	0.394	0.095
	(0.296)	(0.284)	(0.287)	(0.287)
Share of home-based patients	0.043	0.040	0.019	0.046
	(0.039)	(0.040)	(0.044)	(0.038)
Total patient days / 1000		-0.000		
		(0.001)		
Total patient days / 1000 squared		-0.000		
		(0.000)		
Total Visits / 10,000			-0.045***	
			(0.010)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.023*	-0.014	0.020	-0.022*
	(0.010)	(0.012)	(0.012)	(0.010)
Inpatient unit services				-0.071
				(0.039)
Pediatric program services				-0.047
				(0.048)
Bereavement services				0.060**
				(0.021)
Adult day care services				0.245**
				(0.083)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs. Standard errors clustered at county-level in parentheses.

H Regression on individual cost components

The controls in the following regressions are: share of patients with cancer, share of patients with length of stay less than or equal to 7 days, average actual length of stay, share of non-routine care days, share of patients served at home, entry in 2002 indicator, entry time measure, log of operating costs, for-profit status, lagged county-level-mean slope of per-visit costs (and its square) up to two lags.

	Drug, biologicals, and infusion
log(firm count)	-5.176*** (0.770)
For-profit	0.446 (0.788)
Entry time	0.128 (0.115)
Entry before 2002	1.294 (0.876)
County Fixed Effects	Yes
Controls	Yes

	Durable medical equipment
log(firm count)	-1.828* (0.794)
For-profit	-0.080 (0.612)
Entry time	0.076 (0.056)
Entry before 2002	-0.211 (0.768)
County Fixed Effects	Yes
Controls	Yes

Patient transportation	
log(firm count)	0.048 (0.105)
For-profit	0.115 (0.095)
Entry time	0.003 (0.011)
Entry before 2002	0.172 (0.089)
County Fixed Effects	Yes
Controls	Yes

Imaging, lab, and diagnostics	
log(firm count)	0.029 (0.072)
For-profit	0.036 (0.029)
Entry time	-0.003 (0.007)
Entry before 2002	-0.056 (0.053)
County Fixed Effects	Yes
Controls	Yes

Medical supplies	
log(firm count)	0.100 (0.423)
For-profit	0.070 (0.473)
Entry time	0.190*** (0.044)
Entry before 2002	1.168* (0.444)
County Fixed Effects	Yes
Controls	Yes

Outpatient services	
log(firm count)	1.877 (1.888)
For-profit	-0.442 (0.434)
Entry time	-0.132 (0.118)
Entry before 2002	-0.036 (0.219)
County Fixed Effects	Yes
Controls	Yes

Physician services	
log(firm count)	1.892** (0.660)
For-profit	-0.693 (0.563)
Entry time	0.139* (0.063)
Entry before 2002	-0.941 (0.505)
County Fixed Effects	Yes
Controls	Yes

Nursing care	
log(firm count)	14.870*** (3.425)
For-profit	-13.290*** (3.202)
Entry time	-1.783*** (0.402)
Entry before 2002	-0.984 (3.451)
County Fixed Effects	Yes
Controls	Yes

Rehabilitation services expenses	
log(firm count)	-3.105 (2.868)
For-profit	-0.449 (0.314)
Entry time	0.070 (0.120)
Entry before 2002	0.340 (0.791)
County Fixed Effects	Yes
Controls	Yes

Medical social services expenses	
log(firm count)	3.048 (2.605)
For-profit	-3.803** (1.315)
Entry time	-0.025 (0.203)
Entry before 2002	3.418 (1.959)
County Fixed Effects	Yes
Controls	Yes

Spiritual counseling	
log(firm count)	0.406 (0.581)
For-profit	-0.167 (0.327)
Entry time	-0.060 (0.050)
Entry before 2002	-0.563 (0.372)
County Fixed Effects	Yes
Controls	Yes

Dietary counseling	
log(firm count)	0.030 (0.103)
For-profit	-0.053 (0.038)
Entry time	-0.009 (0.005)
Entry before 2002	-0.040 (0.044)
County Fixed Effects	Yes
Controls	Yes

Counseling services (other)	
log(firm count)	0.470 (0.565)
For-profit	-0.473 (0.276)
Entry time	0.042 (0.027)
Entry before 2002	0.555 (0.508)
County Fixed Effects	Yes
Controls	Yes

Home health aides and homemakers	
log(firm count)	0.417 (0.948)
For-profit	-0.183 (0.770)
Entry time	-0.140 (0.072)
Entry before 2002	-0.736 (0.749)
County Fixed Effects	Yes
Controls	Yes

	Other visiting services
log(firm count)	3.226* (1.423)
For-profit	-4.388** (1.395)
Entry time	-0.099 (0.193)
Entry before 2002	-1.080 (1.719)
County Fixed Effects	Yes
Controls	Yes

	Other program costs
log(firm count)	-1.928 (3.313)
For-profit	1.364 (1.500)
Entry time	-0.299 (0.314)
Entry before 2002	-3.263 (2.443)
County Fixed Effects	Yes
Controls	Yes

I Useful references and links from the palliative care literature

Some papers that helped inform this paper are [Gibson et al. \(2013\)](#) and [Huskamp et al. \(2008\)](#), among others.

Useful links:

1. <https://www.milliman.com/en/insight/hospice-medicare-margins-analysis-of-patient-and-hospice-characteristics-utilization-analysis>
2. https://www.medpac.gov/wp-content/uploads/import_data/scrape_files/docs/default-source/reports/Mar10_Ch02E_APPENDIX.pdf
3. https://allianceforcareathome.org/wp-content/uploads/NHPCO_Hospice_Staffing_Framework.pdf
4. https://allianceforcareathome.org/wp-content/uploads/PC_Staffing_grab-go.pdf
5. https://www.medpac.gov/wp-content/uploads/import_data/scrape_files/docs/default-source/contractor-reports/the-medicare-hospice-payment-system-a-preliminary-consideration-of-potential-refinements.pdf

J Entry restrictions and fraud in California hospice industry

The following lists links to reports from the California state government and various news sources about fraud and entry restrictions in the California hospice industry.

In general, the worry is that the number of hospices in California (and mainly LA) has grown extremely rapidly. These reports find suspicious patterns about hospices in LA regarding their location. The California auditor and DOJ suspect that some/many of these hospices are fake and bill Medicare without rendering any services. Others convince non-terminally ill patients to enroll, or convince patients to take on services that they have to later pay for out of pocket. To combat this, California has introduced entry restrictions for new hospices (CON laws), which have been extended multiple times.

While there have been accusations of widespread fraud in LA, the extent of it is unclear. There have been a few high-profile prosecutions where fake hospices billed Medicare for millions of dollars, but it is unclear how widespread this is. The California DOJ office reports filing cases on this matter. The investigations are still ongoing at the time of this writing so this may change with time.

1. Entry restrictions (initial): <https://hospicenews.com/2021/10/05/california-passes-major-hospice-reform-laws/>
2. Entry restrictions (extension till 2024): <https://hospicenews.com/2022/12/01/new-california-law-further-tightens-hospice-license-oversight/>
3. Entry restrictions (extension till 2027): <https://www.cdph.ca.gov/Programs/CHCQ/LCP/Pages/AFL-25-04.aspx>
4. Report from California State Auditor: <https://information.auditor.ca.gov/reports/2021-123/index.html>

5. Press release by California Attorney General: https://oag.ca.gov/news/press-releases/confronting-hospice-fraud-attorney-general-bonta-launches-public-awareness?utm_medium=email&utm_source=govdelivery

K Robustness checks for main results

K.1 With number-equivalent HHI

Table 34: Regression of per-visit costs on competition, for-profit status, and entry time. Competition measured as number-equivalent HHI calculated using within-hospice market share.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
Total patient days / 1000		0.002 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.040** (0.011)	
Total Visits / 10,000 squared			0.000 (0.000)	
log(operating cost)	-0.007 (0.008)	-0.012 (0.010)	0.034* (0.013)	-0.014 (0.008)
Inpatient unit services				-0.034 (0.045)
Pediatric program services				0.020 (0.052)
Bereavement services for non-hospice survivors				0.090*** (0.025)
Adult day care services				0.269* (0.100)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 35: Regression of per-visit costs on competition, for-profit status, and entry time. Competition measured as number-equivalent HHI calculated using within-hospice market share.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(Number-equivalent HHI)	0.053 (0.048)	0.057 (0.050)	0.071 (0.056)	0.053 (0.047)
For-profit	-0.187*** (0.045)	-0.189*** (0.046)	-0.187*** (0.042)	-0.187*** (0.047)
Entry time	-0.116*** (0.029)	-0.119*** (0.028)	-0.138*** (0.027)	-0.110*** (0.028)
Entry before 2002	-0.097* (0.048)	-0.099* (0.047)	-0.116* (0.045)	-0.102* (0.048)
Total patient days / 1000		-0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.044*** (0.011)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.031** (0.011)	-0.022 (0.012)	0.010 (0.013)	-0.032** (0.010)
Inpatient unit services				-0.056 (0.037)
Pediatric program services				-0.026 (0.047)
Bereavement services				0.057** (0.021)
Adult day care services				0.261* (0.099)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 36: First stage of IV using number equivalent HHI.

	log(Number-equivalent HHI)			
	(1)	(2)	(3)	(4)
log(market size)	3.311*** (0.481)	3.301*** (0.479)	3.326*** (0.480)	3.304*** (0.476)
Share of patients w/ cancer	0.062 (0.054)	0.069 (0.055)	0.060 (0.052)	0.067 (0.059)
Share of patients w/ <7 days LOS	-0.029 (0.086)	-0.031 (0.085)	-0.020 (0.085)	-0.027 (0.085)
Average length of stay	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Share of non-routine care days	-0.245 (0.207)	-0.194 (0.209)	-0.158 (0.187)	-0.235 (0.196)
Share of home-based patients	-0.036 (0.026)	-0.036 (0.026)	-0.041 (0.026)	-0.039 (0.025)
For-profit	0.038 (0.020)	0.039 (0.020)	0.038 (0.020)	0.036* (0.018)
Entry time	0.022* (0.009)	0.023* (0.009)	0.022* (0.009)	0.022* (0.009)
Entry before 2002	0.096 (0.065)	0.102 (0.064)	0.097 (0.065)	0.101 (0.066)
Total patient days / 1000		0.001 (0.001)		
Total patient days / 1000 squared		-0.000** (0.000)		
Total Visits / 10,000			-0.010 (0.006)	
Total Visits / 10,000 squared			0.000 (0.000)	
log(operating cost)	0.009 (0.008)	0.010 (0.007)	0.019 (0.010)	0.009 (0.008)
Inpatient unit services				-0.031 (0.026)
Pediatric program services				0.015 (0.027)
Bereavement services				-0.016 (0.011)
Adult day care services				-0.081 (0.132)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of logs of market-level-mean per-visit costs.

K.2 Limited to markets with fewer than 50 firms

Table 37: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 50 firms.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	-0.125*** (0.023)	-0.125*** (0.024)	-0.127*** (0.028)	-0.102*** (0.022)
Total patient days / 1000		0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.041** (0.012)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.011 (0.010)	-0.012 (0.012)	0.033* (0.015)	-0.020* (0.009)
Inpatient unit services				-0.018 (0.045)
Pediatric program services				0.021 (0.052)
Bereavement services for non-hospice survivors				0.091** (0.028)
Adult day care services				0.272* (0.104)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,766	2,766	2,766	2,766

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 38: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 50 firms.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.057 (0.042)	0.059 (0.042)	0.064 (0.050)	0.057 (0.041)
For-profit	-0.190*** (0.047)	-0.193*** (0.047)	-0.189*** (0.043)	-0.191*** (0.048)
Entry time	-0.015** (0.005)	-0.016** (0.005)	-0.016*** (0.004)	-0.015** (0.005)
Entry before 2002	-0.026 (0.040)	-0.026 (0.040)	-0.022 (0.041)	-0.036 (0.040)
Total patient days / 1000		-0.001 (0.001)		
Total patient days / 1000 squared		0.000 (0.000)		
Total Visits / 10,000			-0.045*** (0.011)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.035** (0.011)	-0.023 (0.014)	0.011 (0.014)	-0.038*** (0.010)
Inpatient unit services				-0.048 (0.040)
Pediatric program services				-0.024 (0.048)
Bereavement services				0.059* (0.023)
Adult day care services				0.269** (0.093)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,766	2,766	2,766	2,766

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 39: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 50 firms.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.075 (0.044)	0.077 (0.044)	0.089 (0.054)	0.075 (0.043)
For-profit	-0.162** (0.050)	-0.164** (0.050)	-0.164** (0.048)	-0.163** (0.051)
Entry time	-0.015** (0.005)	-0.015** (0.004)	-0.016*** (0.004)	-0.014** (0.004)
Entry before 2002	-0.028 (0.043)	-0.027 (0.043)	-0.026 (0.043)	-0.038 (0.044)
Share of patients w/ cancer	0.239 (0.126)	0.245 (0.127)	0.234 (0.127)	0.254* (0.122)
Share of patients w/ <7 days LOS	-0.199 (0.120)	-0.193 (0.118)	-0.154 (0.115)	-0.214 (0.119)
Average length of stay	-0.002** (0.001)	-0.002** (0.001)	-0.002** (0.001)	-0.002** (0.001)
Share of non-routine care days	-0.030 (0.316)	0.056 (0.304)	0.326 (0.304)	0.026 (0.308)
Share of home-based patients	0.073 (0.047)	0.071 (0.048)	0.050 (0.051)	0.077 (0.045)
Total patient days / 1000		-0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.046*** (0.011)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.027* (0.011)	-0.017 (0.014)	0.017 (0.015)	-0.029* (0.011)
Inpatient unit services				-0.060 (0.040)
Pediatric program services				-0.036 (0.049)
Bereavement services				0.063** (0.023)
Adult day care services				0.274** (0.094)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,766	2,766	2,766	2,766

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

K.3 Limited to markets with fewer than 30 firms

Table 40: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 30 firms.

Table 41: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	-0.115*** (0.031)	-0.116*** (0.031)	-0.113** (0.032)	-0.089** (0.030)
Total patient days / 1000		0.000 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.044** (0.014)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.014 (0.009)	-0.013 (0.014)	0.033* (0.016)	-0.024** (0.009)
Inpatient unit services				-0.014 (0.045)
Pediatric program services				0.020 (0.053)
Bereavement services for non-hospice survivors				0.097** (0.030)
Adult day care services				0.287** (0.098)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,461	2,461	2,461	2,461

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 42: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 30 firms.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.101*	0.106*	0.115*	0.102*
	(0.049)	(0.049)	(0.056)	(0.047)
For-profit	-0.198***	-0.202***	-0.198***	-0.198***
	(0.049)	(0.048)	(0.044)	(0.050)
Entry time	-0.016*	-0.017**	-0.018**	-0.016**
	(0.006)	(0.005)	(0.005)	(0.006)
Entry before 2002	-0.014	-0.015	-0.012	-0.027
	(0.044)	(0.043)	(0.045)	(0.045)
Total patient days / 1000		-0.002		
		(0.001)		
Total patient days / 1000 squared		0.000		
		(0.000)		
Total Visits / 10,000			-0.048***	
			(0.013)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.040***	-0.023	0.011	-0.043***
	(0.011)	(0.016)	(0.016)	(0.010)
Inpatient unit services				-0.044
				(0.041)
Pediatric program services				-0.030
				(0.048)
Bereavement services				0.062*
				(0.026)
Adult day care services				0.283**
				(0.086)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,461	2,461	2,461	2,461

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 43: Regression of per-visit costs on competition, for-profit status, and entry time. Limited to markets with fewer than 30 firms.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.118*	0.123*	0.142*	0.119*
	(0.051)	(0.052)	(0.061)	(0.050)
For-profit	-0.172**	-0.176**	-0.175***	-0.173**
	(0.052)	(0.051)	(0.049)	(0.052)
Entry time	-0.016*	-0.016**	-0.017**	-0.015*
	(0.006)	(0.005)	(0.005)	(0.006)
Entry before 2002	-0.014	-0.014	-0.015	-0.027
	(0.047)	(0.047)	(0.048)	(0.049)
Share of patients w/ cancer	0.225	0.234	0.230	0.246*
	(0.123)	(0.125)	(0.124)	(0.120)
Share of patients w/ <7 days LOS	-0.224	-0.211	-0.155	-0.241
	(0.140)	(0.138)	(0.134)	(0.142)
Average length of stay	-0.002**	-0.002*	-0.002*	-0.002**
	(0.001)	(0.001)	(0.001)	(0.001)
Share of non-routine care days	-0.084	0.017	0.281	-0.031
	(0.342)	(0.331)	(0.329)	(0.329)
Share of home-based patients	0.052	0.049	0.026	0.055
	(0.063)	(0.063)	(0.065)	(0.061)
Total patient days / 1000		-0.001		
		(0.001)		
Total patient days / 1000 squared		0.000		
		(0.000)		
Total Visits / 10,000			-0.049***	
			(0.013)	
Total Visits / 10,000 squared			0.001	
			(0.000)	
log(operating cost)	-0.033*	-0.019	0.016	-0.035**
	(0.012)	(0.016)	(0.016)	(0.012)
Inpatient unit services				-0.055
				(0.041)
Pediatric program services				-0.042
				(0.048)
Bereavement services				0.066*
				(0.026)
Adult day care services				0.287**
				(0.085)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,461	2,461	2,461	2,461

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

K.4 Results with LA county included

Table 44: Regression of per-visit costs on competition and controls.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	-0.090** (0.027)	-0.089** (0.028)	-0.094** (0.028)	-0.084*** (0.023)
Total patient days / 1000		0.002 (0.002)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.041*** (0.010)	
Total Visits / 10,000 squared			0.001* (0.000)	
log(operating cost)	0.000 (0.006)	-0.010 (0.006)	0.037*** (0.009)	-0.003 (0.007)
Inpatient unit services				-0.056 (0.030)
Pediatric program services				-0.015 (0.050)
Bereavement services for non-hospice survivors				0.071*** (0.018)
Adult day care services				0.191 (0.160)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	5,085	5,085	5,085	5,085

Notes: IV regression of log(per-visit cost) on log(firm count) with county fixed effects and controls. Standard errors clustered at county-level in parentheses. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 45: Grouped-quantile regressions of per-visit costs on competition.

Quantile	Coefficient	S.E.	Quantile	Coefficient	S.E.
0.1	-43.3189	13.6307	0.1	-50.066	15.4714
0.2	-37.2306	13.3458	0.2	-42.3478	15.3888
0.3	-32.8515	13.1767	0.3	-36.2643	15.4396
0.4	-28.7941	13.0962	0.4	-31.0552	15.5433
0.5	-22.56	13.1887	0.5	-24.7463	15.7815
0.6	-15.1261	13.2179	0.6	-15.8003	15.8384
0.7	-8.18223	13.5214	0.7	-6.39695	16.2494
0.8	-1.34242	14.0879	0.8	2.47288	17.022
0.9	6.66969	14.9671	0.9	11.5009	18.1128

(a)
(b)

Notes: Grouped-quantile IV regression of per-visit costs on competition, county fixed effects, and semi-parametric order-statistic bias correction. Table 8b also includes second-order polynomials of one-year-lag and two-year-lag of county-level-mean of per-visit costs. Robust standard errors are reported under S.E. See Section A.1 for details about implementation.

Table 46: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.022 (0.020)	0.021 (0.021)	0.033 (0.022)	0.018 (0.020)
For-profit	-0.158*** (0.033)	-0.157*** (0.033)	-0.161*** (0.031)	-0.163*** (0.033)
Entry time	-0.012** (0.004)	-0.011** (0.004)	-0.014*** (0.004)	-0.011** (0.003)
Entry before 2002	-0.001 (0.031)	0.002 (0.032)	0.002 (0.033)	-0.007 (0.032)
Total patient days / 1000		0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.048*** (0.009)	
Total Visits / 10,000 squared			0.001* (0.000)	
log(operating cost)	-0.020* (0.008)	-0.020** (0.007)	0.020* (0.008)	-0.019* (0.009)
Inpatient unit services				-0.071** (0.025)
Pediatric program services				-0.056 (0.046)
Bereavement services				0.043** (0.014)
Adult day care services				0.184 (0.160)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	5,085	5,085	5,085	5,085

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

Table 47: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	0.030 (0.020)	0.029 (0.021)	0.044 (0.024)	0.026 (0.021)
For-profit	-0.135*** (0.032)	-0.134*** (0.032)	-0.137*** (0.031)	-0.141*** (0.032)
Entry time	-0.012*** (0.003)	-0.011*** (0.003)	-0.014*** (0.003)	-0.011*** (0.003)
Entry before 2002	-0.006 (0.033)	-0.002 (0.033)	-0.004 (0.035)	-0.011 (0.034)
Share of patients w/ cancer	0.185* (0.070)	0.187* (0.070)	0.163* (0.070)	0.197** (0.070)
Share of patients w/ <7 days LOS	-0.118* (0.054)	-0.119* (0.054)	-0.094 (0.054)	-0.122* (0.054)
Average length of stay	-0.000 (0.000)	-0.000 (0.000)	-0.001* (0.000)	-0.000 (0.000)
Share of non-routine care days	-0.137 (0.111)	-0.126 (0.110)	0.007 (0.147)	-0.101 (0.110)
Share of home-based patients	0.134** (0.039)	0.134** (0.038)	0.127** (0.046)	0.135** (0.038)
Total patient days / 1000		0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.047*** (0.010)	
Total Visits / 10,000 squared			0.001* (0.000)	
log(operating cost)	-0.012 (0.008)	-0.013* (0.006)	0.027** (0.008)	-0.011 (0.008)
Inpatient unit services				-0.072** (0.025)
Pediatric program services				-0.066 (0.046)
Bereavement services				0.043** (0.015)
Adult day care services				0.196 (0.164)
County Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	5,085	5,085	5,085	5,085

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials ~~100~~ logs of market-level-mean per-visit costs.

Table 48: Regression of per-visit costs on competition, for-profit status, and entry time.

	log(per-visit cost)			
	(1)	(2)	(3)	(4)
log(firm count)	-0.027 (0.062)	-0.027 (0.063)	-0.031 (0.073)	-0.031 (0.062)
For-profit	-0.192*** (0.047)	-0.194*** (0.047)	-0.193*** (0.043)	-0.194*** (0.048)
Entry time	-0.014*** (0.004)	-0.014*** (0.003)	-0.015*** (0.003)	-0.013*** (0.003)
Entry before 2002	-0.006 (0.042)	-0.004 (0.042)	-0.003 (0.043)	-0.015 (0.043)
Total patient days / 1000		-0.001 (0.001)		
Total patient days / 1000 squared		-0.000 (0.000)		
Total Visits / 10,000			-0.043*** (0.011)	
Total Visits / 10,000 squared			0.001 (0.000)	
log(operating cost)	-0.034*** (0.010)	-0.026* (0.012)	0.007 (0.012)	-0.035*** (0.010)
Inpatient unit services				-0.054 (0.038)
Pediatric program services				-0.036 (0.048)
Bereavement services				0.056* (0.021)
Adult day care services				0.272** (0.099)
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	3,114	3,114	3,114	3,114

Notes: Standard errors clustered at county-level in parentheses. Entry time is measured as number of years since 2002. Controls are second-order polynomials of lags of market-level-mean per-visit costs.

L Descriptive statistics and robustness checks for instrumental variables

L.1 Robustness checks for exclusion restriction

Table 49: Regression of per-visit costs on population density, firm fixed effects, and controls.

	log(per-visit cost)		
	(1)	(2)	(3)
log(population density)	-0.141 (0.189)	0.147 (0.206)	0.188 (0.281)
log(operating cost)		-0.054** (0.015)	-0.057** (0.016)
Mean per-visit cost _{t-1}			0.005* (0.002)
Mean per-visit cost _{t-2}			-0.001 (0.002)
Mean per-visit cost _{t-1} ²			-0.000 (0.000)
Mean per-visit cost _{t-2} ²			0.000 (0.000)
Firm Fixed Effects	Yes	Yes	Yes
Obs.	3,226	3,226	3,022

Notes: Standard errors clustered at county-level in parentheses.

Table 50: Regression of per-visit costs on market size, firm fixed effects, and controls.

	log(per-visit cost)		
	(1)	(2)	(3)
log(market size)	-0.148 (0.140)	0.038 (0.148)	0.117 (0.173)
log(operating cost)		-0.052** (0.015)	-0.056*** (0.016)
Mean per-visit cost _{t-1}			0.005* (0.002)
Mean per-visit cost _{t-2}			-0.000 (0.002)
Mean per-visit cost _{t-1} ²			-0.000 (0.000)
Mean per-visit cost _{t-2} ²			0.000 (0.000)
Firm Fixed Effects	Yes	Yes	Yes
Obs.	3,226	3,226	3,022

Notes: Standard errors clustered at county-level in parentheses.

L.2 Descriptive statistics for expected deaths

Figure 20: Histograms of expected deaths and $\log(\text{expected deaths})$ across market-years.

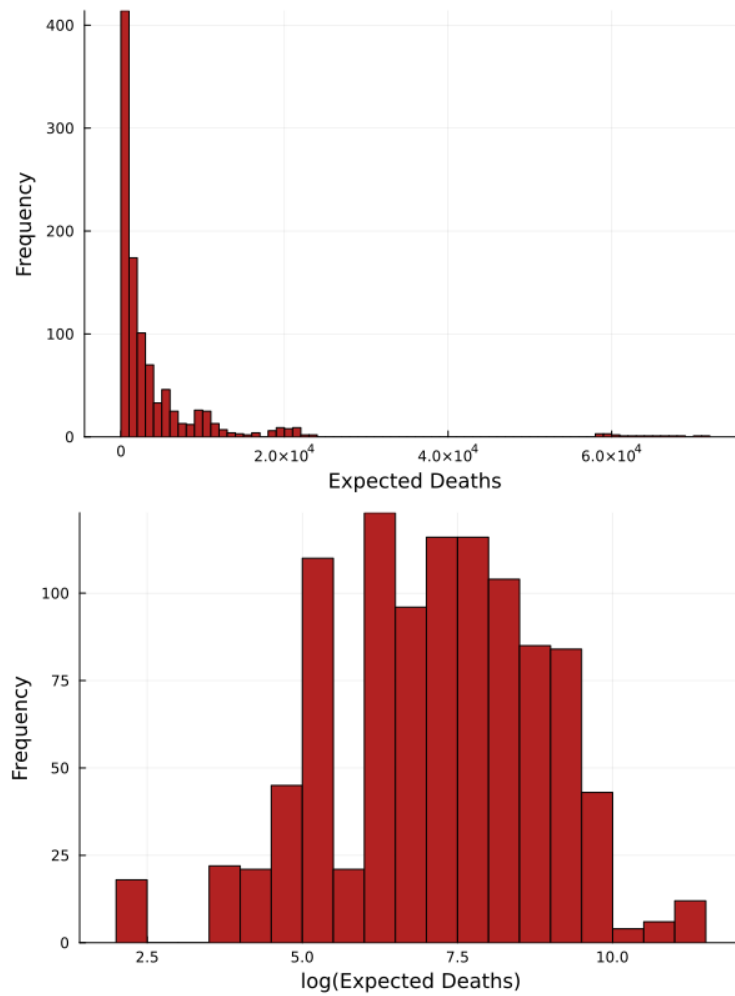


Figure 21: Evolution of expected deaths for a subset of counties.

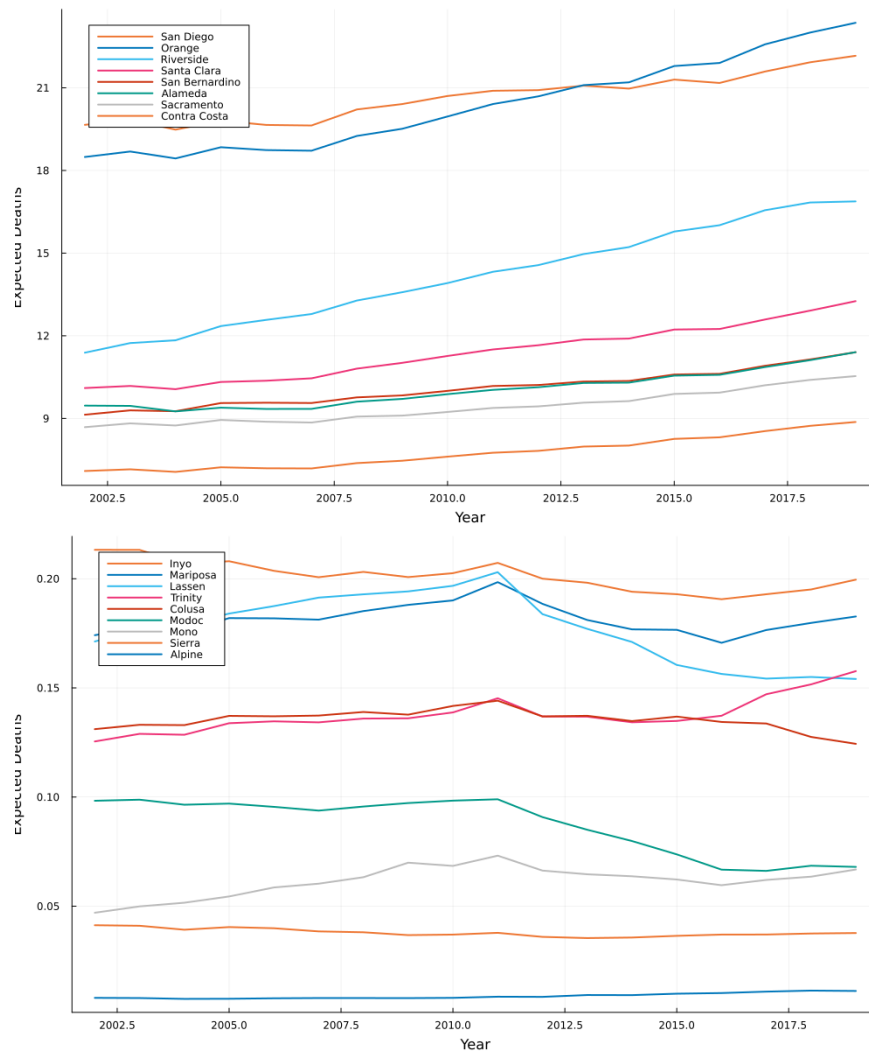


Figure 22: Evolution of log of expected deaths for a subset of counties.

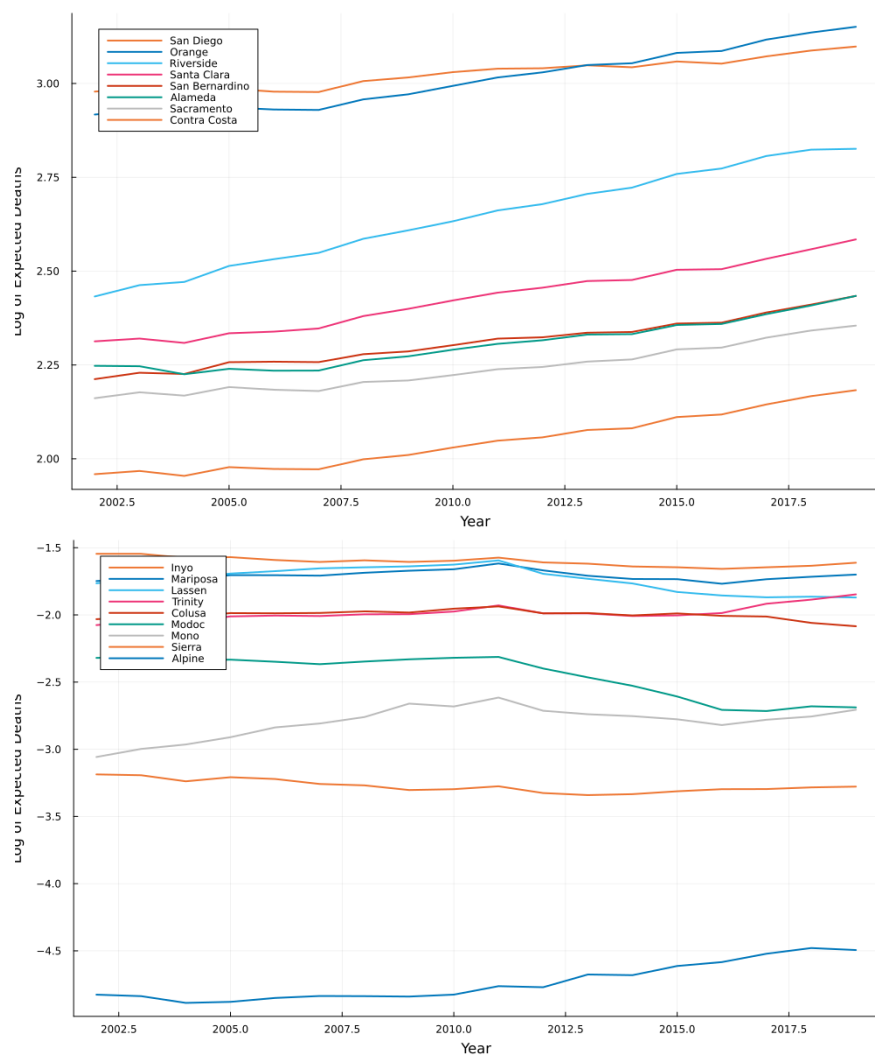


Figure 23: Residualized expected deaths: histogram and evolution for a subset of counties.

